Estimating the sampling distribution of the maximum likelihood estimator of the parameters of a series system given a sample of masked system failure times

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Abstract

First, the *parametric family* of an abstract *series system* is derived. Then, the *asymptotic sampling distribution* (*multivariate normal*) of a *maximum likelihood estimator* of the true parameter is derived conditioned on the *information* given by a random sample of *masked system failure times*. The asymptotic sampling distributions of statistics that are functions of the parameter follow as a result. Finally, these results are applied to series systems in which component lifetimes are *exponentially* distributed, which lead to closed formulas on a set of *minimally sufficient statistics*.

Contents

1 Introduction

We consider series systems consisting of some fixed number of components with uncertain lifetimes. We desire a mathematical model of the system that may be used to predict the failure time of the system and its component cause. However, since the system lifetime and the component cause of failure is uncertain, we are interested in probabilistically modeling the system lifetime, e.g., there is a 75% chance that component 3 will cause a system failure in the next 3 years.

We assume the system belongs to some parametric family but whose true parameter index is unknown. Furthermore, we also assume there is a sample of *masked system failure times* that carry information about the true parameter index.

Using the information in the sample, we employ the frequentist approach of repeated experiments which induces a sampling distribution on the maximum likelihood point estimator of the fixed but unknown true parameter index where *a priori* any value in the parameter space is equally likely. The point estimator converges to a multivariate normal sampling distribution with a mean given by the true parameter index and a variance that is inversely proportional to sample size.

2 Probabilistic model

A probabilistic model is specified by equations involving random variables which make assumptions about how observable data about the system is generated. In what follows, we describe our model and observable data.

2.1 Series system

The system consists of m components. We make the following assumption about the state of the components.

Assumption 1. *The j^{th} component is initially in a non-failed state and permanently enters a failed state at some uncertain time $t_j > 0$ for $j = 1, \ldots, m$.*

Components have uncertain lifetimes as given by the following definition.

Definition 2.1. *The lifetime of the j^{th} component is given by the continuous random variable $\mathrm{T}_j \in (0, \infty)$ for $j = 1, \ldots, m$.*

We make the following assumption about the joint distribution of the component lifetimes.

Assumption 2. *The random variables $\mathrm{T}_1, \ldots, \mathrm{T}_m$ are mutually independent.*

The system is a series system.

Assumption 3. *The system is in a non-failed state if all m components are in a non-failed state, otherwise the system is in a failed state.*

A system in which at least k of the components must be in a non-failed state for the system to be in a non-failed system is denoted a k-out-of-m system, thus a series system is an m-out-of-m system.

The uncertain lifetime of the system is given by the following definition.

Theorem 2.1. *The lifetime of the system is given by the random variable*

$$\mathrm{S} = \min(\mathrm{T}_1, \ldots, \mathrm{T}_m) \in (0, \infty) \,, \tag{2.1}$$

where $\mathrm{T}_1, \ldots, \mathrm{T}_m$ are the component lifetimes.

Proof. A series system fails whenever any component fails, therefore its lifetime is equal to the lifetime of the component with the minimum lifetime. □

2.2 Probability distribution functions and parametric families

As random variables, it is not certain which values $\mathrm{S}, \mathrm{T}_1, \ldots, \mathrm{T}_m$ will realize. However, their respective probability distribution functions quantify the uncertainty. The cumulative distribution function is given by the following definition.

Definition 2.2. *Let X be some random variable. The probability that $\mathrm{X} \leq x$ is given by its cumulative distribution function, denoted by*

$$F_{\mathrm{X}}(x) = \Pr[\mathrm{X} \leq x] \,. \tag{2.2}$$

The probability density function is given by the following definition.

Definition 2.3. *Let* X *be some continuous random variable. The* relative likelihood *that* $\mathrm{X} = x$ *is given by its probability density function, denoted by*

$$f_{\mathrm{X}}(x) = \frac{\mathrm{d}}{\mathrm{d}t} F_{\mathrm{X}}(t)\,. \tag{2.3}$$

By the Second Fundamental Theorem of Calculus, the probability that a continuous random variable $\mathrm{X} \in (a, b]$, $a \leq b$, is given by

$$\Pr[a < \mathrm{X} \leq b] = F_{\mathrm{X}}(b) - F_{\mathrm{X}}(a) = \int_a^b f_{\mathrm{X}}(s) \mathrm{d}s\,. \tag{2.4}$$

By these definitions, the relative likelihood that the system will fail at time t is $f_{\mathrm{S}}(t)$ and probability the j^{th} component will before time t is $F_{\mathrm{T}_j}(t)$.

Notation. We let $F_j \equiv F_{\mathrm{T}_j}$ and $f_j \equiv f_{\mathrm{T}_j}$. Column vectors are denoted by boldface lower case letters, e.g., $\mathbf{x}$, and row vectors are denoted by the transpose of column vectors, e.g., $\mathbf{x}^{\mathsf{T}}$. Matrices are denoted by boldface upper case greek letters, e.g., $\boldsymbol{A}$. The j^{th} column of $\boldsymbol{A}$ is denoted by $\boldsymbol{A}_j$ (or $[\boldsymbol{A}]_j$) and the (j, k)-th element of $\boldsymbol{A}$ is denoted by $\boldsymbol{A}_{j\,k}$ (or $[\boldsymbol{A}]_{j\,k}$).

We restrict our attention to parametrized families of probability distribution functions. The system lifetime has a probability density function that is a member of the following parametric family.

Definition 2.4. *The system lifetime has a probability density function that is a member of the parametric family of probability density functions denoted by*

$$\mathcal{F}_{\boldsymbol{\Theta}} = \left\{ f_{\mathrm{S}}\left(\,\cdot \mid \boldsymbol{\Theta} \right) : \boldsymbol{\Theta} \in \mathbb{R}^{q \times m} \right\}\,, \tag{2.5}$$

where $\boldsymbol{\Theta}_j \in \boldsymbol{\Omega}$ *corresponds to the parameter index of the* j^{th} *component lifetime.*

Particular values of $\boldsymbol{\Theta}$ index particular probability density functions within the $\mathcal{F}_{\boldsymbol{\Theta}}$-family. The true probability density function of the system lifetime is given by the following definition.

Definition 2.5. *The distribution of the system lifetime has a true parameter index denoted by* $\boldsymbol{\Theta}^\star$*, and we say that*

$$\mathrm{S} \sim f_{\mathrm{S}}(\,\cdot \mid \boldsymbol{\Theta}^\star)\,. \tag{2.6}$$

By these definitions, it follows that

$$\mathrm{T}_j \sim f_j\left(\,\cdot \mid \boldsymbol{\Theta}_j^\star\right)\,. \tag{2.7}$$

2.3 Masked system failure times

By theorem 2.1, if a system failure occurs at time t, then one of the components failed at time t (only a single component can cause a system failure since two or more continuous random variables cannot realize the same value).

Definition 2.6. *The discrete random variable indicating the component responsible for a system failure is denoted by* K *with a sample space* $\mathsf{K} = \{1, \ldots, m\}$*. That is,* $\mathrm{K} = j$ *indicates that* $\mathrm{T}_j < \mathrm{T}_k$ *for* $k \in \mathsf{K} \smallsetminus \{j\}$*.*

We may not be certain which component caused a system failure, but may be able to narrow the component cause to some *subset*. The set of all component sets is given by the power set of K, denoted by $\mathcal{P}(\mathsf{K})$.

The probability that a component in a set $\mathsf{A} \in \mathcal{P}(\mathsf{K})$ is the cause of a system failure follows from the definition of K, i.e., $\Pr[\mathrm{K} \in \mathsf{A}]$.

The total probability over $\mathcal{P}(\mathsf{K})$ does not sum to unity since component sets are not disjoint events with respect to the sample space of K. We consider another random variable (random set) in which the sample space is defined by the outcomes in $\mathcal{P}(\mathsf{K})$.

The uncertain component sets are given by the following definition.

Definition 2.7. *Let $\mathcal{C}$ denote the random set of components that are the most likely causes of system failure. The sample space of $\mathcal{C}$ is $\mathcal{P}(\mathsf{K})$.*

The *cardinality* of a component set could theoretically convey information about the system. As a simplification, we make the following assumption.

Assumption 4. *The cardinality of the random set $\mathcal{C}$, denoted by the discrete random variable*

$$\mathrm{W} = |\mathcal{C}|\,, \tag{2.8}$$

is independent of the random system lifetime S *and the random component cause $\mathcal{C}$ and therefore does not carry information about the true parameter index $\boldsymbol{\Theta}^\star$.*

Definition 2.8. *A* masked system failure time *consists of a system failure time* t *and a corresponding set of components* C *that* plausibly *caused the system failure.*

Assumption 5. *A random* masked system failure time *is a jointly distributed random system lifetime* S *and random component set $\mathcal{C}$.*

Assumption 6. *A random sample of n* masked system failure times*, denoted by*

$$\mathbf{M_n} = (\mathrm{S}_1, \mathcal{C}_1)\ , \ \dots\ , \ (\mathrm{S}_n, \mathcal{C}_n)\ , \tag{2.9}$$

consists of n independent and identically jointly distributed pairs of system failure times and candidate sets.

Assumption 7. *The only information we have about the system and its true parameter index $\boldsymbol{\Theta}^\star$ is given by observing a random sample of n masked system failure times,*

$$(\mathrm{S}_1 = \mathrm{t}_1, \mathcal{C}_1 = \mathsf{C}_1)\,, \dots, (\mathrm{S}_n = \mathrm{t}_\mathrm{n}, \mathcal{C}_\mathrm{n} = \mathsf{C}_\mathsf{n})\ . \tag{2.10}$$

Given the assumed model, the object of statistical interest is the (unknown) true parameter index $\boldsymbol{\Theta}^\star$. Provided an estimate of $\boldsymbol{\Theta}^\star$, any characteristic that is a function of $\boldsymbol{\Theta}^\star$ can be estimated as a result, thus the primary objective is to use the *information* in a sample of n masked system failure times to estimate $\boldsymbol{\Theta}^\star$ with some quantifiable uncertainty.

3 Derivation of parametric probability distribution functions

Chapter 2 described the probabilistic model. In what follows, we derive the parametric functions that are entailed by the model.

3.1 System lifetime

The complement of the cumulative distribution function is the reliability function and is given by the following definition.

Definition 3.1. *Let* X *be some random variable. The probability that* $\mathrm{X} > x$ *is given by its reliability function, denoted by*

$$R_{\mathrm{X}}(x) = \Pr\left[\mathrm{X} > x\right] = 1 - F_{\mathrm{X}}(x)\ . \tag{3.1}$$

The reliability function of the system lifetime is given by the following theorem.

Theorem 3.1. *The system lifetime has a reliability function given by*

$$R_{\mathrm{S}}\left(\mathrm{t} \mid \Theta^{\star}\right) = \prod_{j=1}^{m} R_j\left(\mathrm{t} \mid \Theta^{\star}_{\mathrm{j}}\right)\ , \tag{3.2}$$

where R_j *is the reliability function of the* j^{th} *component.*

Proof. By definition 3.1, the reliability function is given by

$$R_{\mathrm{S}}\left(\mathrm{t} \mid \Theta^{\star}\right) = \Pr\left[\mathrm{S} > \mathrm{t}\right]\ . \tag{a}$$

By eq. (2.1), $\mathrm{S} = \min(\mathrm{T}_1, \ldots, \mathrm{T}_m)$. Performing this substitution yields

$$R_{\mathrm{S}}(\mathrm{t} \mid \Theta^{\star}) = \Pr[\min(\mathrm{T}_1, \ldots, \mathrm{T}_{\mathrm{m}}) > \mathrm{t}]\,. \tag{b}$$

For the minimum to be larger than t, every component must be larger than t, leading to the equivalent equation

$$R_{\mathrm{S}}(\mathrm{t} \mid \Theta^{\star}) = \Pr[\mathrm{T}_1 > \mathrm{t}, \ldots, \mathrm{T}_{\mathrm{m}} > \mathrm{t}]\,. \tag{c}$$

By assumption 2, the component lifetimes are independent, thus by the axioms of probability

$$R_{\mathrm{S}}\left(\mathrm{t} \mid \Theta^{\star}\right) = \Pr[\mathrm{T}_1 > \mathrm{t}] \cdots \Pr[\mathrm{T}_{\mathrm{m}} > \mathrm{t}]\,. \tag{d}$$

By **??**, the probability that $\mathrm{T}_j > \mathrm{t}$ is equivalent to the reliability function of the j^{th} component lifetime evaluated at t. Performing these substitutions results in

$$R_{\mathrm{S}}\left(t \mid \boldsymbol{\Theta}^{\star}\right) = \prod_{j=1}^{m} R_j(t \mid \boldsymbol{\Theta}^{\star}_j)\,. \tag{e}$$

□

The probability density function of the system's lifetime is given by the following theorem.

Theorem 3.2. *The lifetime of the m-out-of-m system has a probability density function given by*

$$f_{\mathrm{S}}(t \mid \boldsymbol{\Theta}^{\star}) = \sum_{j=1}^{m}\left(f_j(t \mid \boldsymbol{\Theta}^{\star}_j)\prod_{\substack{k=1\\k\neq j}}^{m} R_k(t \mid \boldsymbol{\Theta}^{\star}_k)\right)\ , \tag{3.3}$$

where R_k *and* f_k *are respectively the reliability and probability density functions of the* k^{th} *component.*

Proof. By **??**,

$$f_{\mathrm{S}}\left(t \,|\, \boldsymbol{\Theta}^{\star}\right) = -\frac{\mathrm{d}}{\mathrm{d}t} R_{\mathrm{S}}\left(t \,|\, \boldsymbol{\Theta}^{\star}\right) . \tag{a}$$

By theorem 3.1, this is equivalent to

$$f_{\mathrm{S}}\left(t \,|\, \boldsymbol{\Theta}^{\star}\right) = -\frac{\mathrm{d}}{\mathrm{d}t} \prod_{j=1}^{m} R_j\left(t \,|\, \boldsymbol{\Theta}_j^{\star}\right) . \tag{b}$$

By the product rule, this is equivalent to

$$\begin{aligned} f_{\mathrm{S}}(t) = {} & \frac{\mathrm{d}}{\mathrm{d}t} R_1\left(t \,|\, \boldsymbol{\Theta}_1^{\star}\right) \prod_{j=2}^{m} R_j\left(t \,|\, \boldsymbol{\Theta}_j^{\star}\right) \\ & - R_1\left(t \,|\, \boldsymbol{\Theta}_1^{\star}\right) \frac{\mathrm{d}}{\mathrm{d}t} \prod_{j=2}^{m} R_j\left(t \,|\, \boldsymbol{\Theta}_j^{\star}\right) . \end{aligned} \tag{c}$$

By **??**, we can substitute $-\frac{\mathrm{d}}{\mathrm{d}t} R_1\left(t \,|\, \boldsymbol{\Theta}_1^{\star}\right)$ with $f_1\left(t \,|\, \boldsymbol{\Theta}_1^{\star}\right)$, resulting in

$$\begin{aligned} f_{\mathrm{S}}(t) = {} & f_1(t) \prod_{j=2}^{m} R_j\left(t \,|\, \boldsymbol{\Theta}_j^{\star}\right) \\ & - R_1\left(t \,|\, \boldsymbol{\Theta}_1^{\star}\right) \frac{\mathrm{d}}{\mathrm{d}t} \prod_{j=2}^{m} R_j\left(t \,|\, \boldsymbol{\Theta}_j^{\star}\right) . \end{aligned} \tag{d}$$

Applying the product rule again results in

$$\begin{aligned} f_{\mathrm{S}}(t \,|\, \boldsymbol{\Theta}^{\star}) = {} & f_1\left(t \,|\, \boldsymbol{\Theta}_1^{\star}\right) \prod_{j=2}^{m} R_j\left(t \,|\, \boldsymbol{\Theta}_j^{\star}\right) \\ & + f_2\left(t \,|\, \boldsymbol{\Theta}_2^{\star}\right) \prod_{\substack{j=1 \\ j \neq 2}}^{m} R_j\left(t \,|\, \boldsymbol{\Theta}_j^{\star}\right) \\ & - R_1\left(t \,|\, \boldsymbol{\Theta}_1^{\star}\right) R_2\left(t \,|\, \boldsymbol{\Theta}_2^{\star}\right) \frac{\mathrm{d}}{\mathrm{d}t} \prod_{j=3}^{m} R_j\left(t \,|\, \boldsymbol{\Theta}_j^{\star}\right) . \end{aligned} \tag{e}$$

We see a pattern emerge. Applying the product rule $m-1$ times results in

$$\begin{aligned} f_{\mathrm{S}}\left(t \,|\, \boldsymbol{\Theta}^{\star}\right) = {} & \sum_{j=1}^{m-1} f_j\left(t \,|\, \boldsymbol{\Theta}_j^{\star}\right) \prod_{\substack{k=1 \\ k \neq j}}^{m} R_k\left(t \,|\, \boldsymbol{\Theta}_k^{\star}\right) \\ & - \prod_{j=1}^{m-1} R_j\left(t \,|\, \boldsymbol{\Theta}_j^{\star}\right) \frac{\mathrm{d}}{\mathrm{d}t} R_m\left(t \,|\, \boldsymbol{\Theta}_m^{\star}\right) \\ = {} & \sum_{j=1}^{m} f_j\left(t \,|\, \boldsymbol{\Theta}_j^{\star}\right) \prod_{\substack{k=1 \\ k \neq j}}^{m} R_k\left(t \,|\, \boldsymbol{\Theta}_k^{\star}\right) . \end{aligned} \tag{f}$$

□

The failure rate function simplifies some of the subsequent material and is given by the following definition.

Definition 3.2. *A random variable* X *with a probability density function* f_{X} *and reliability function* R_{X} *has a failure rate function given by*

$$h_{\mathrm{X}}(t) = \frac{f_{\mathrm{X}}(t)}{R_{\mathrm{X}}(t)}\,. \tag{3.4}$$

The failure rate function of the system lifetime is given by the following theorem.

Theorem 3.3. *The system lifetime has a failure rate function given by*

$$h_{\mathrm{S}}(t \mid \boldsymbol{\Theta}^\star) = \sum_{j=1}^{m} h_j(t \mid \boldsymbol{\Theta}_j^\star) \tag{3.5}$$

where h_j *is the failure rate function of the* j^{th} *component lifetime.*

Proof. By eq. (3.4), the failure rate function for the system is given by

$$h_{\mathrm{S}}(t \mid \boldsymbol{\Theta}^\star) = \frac{f_{\mathrm{S}}(t \mid \boldsymbol{\Theta}^\star)}{R_{\mathrm{S}}(t \mid \boldsymbol{\Theta}^\star)}\,. \tag{a}$$

Plugging in these parametric functions results in

$$h_{\mathrm{S}}(t \mid \boldsymbol{\Theta}^\star) = \frac{\sum_{j=1}^{m}\left(f_j(t \mid \boldsymbol{\Theta}_j^\star)\prod_{\substack{k=1\\k\neq j}}^{m} R_k(t \mid \boldsymbol{\Theta}_k^\star)\right)}{\prod_{j=1}^{m} R_j(t \mid \boldsymbol{\Theta}_j^\star)}\,, \tag{b}$$

which can be simplified to

$$h_{\mathrm{S}}(t \mid \boldsymbol{\Theta}^\star) = \sum_{j=1}^{m} f_j(t \mid \boldsymbol{\Theta}_j^\star)/R_j(t \mid \boldsymbol{\Theta}_j^\star) \tag{c}$$

$$= \sum_{j=1}^{m} h_j(t \mid \boldsymbol{\Theta}_j^\star)\,. \tag{d}$$

□

3.2 Component cause of system failure

According to definition 2.6, the component responsible for a system failure is given by the discrete random variable K. The joint likelihood that component k is the cause of a system failure occurring at time t is given by the following theorem.

Theorem 3.4. *The joint probability density function of random variable* K *and random system lifetime* S *is given by*

$$f_{\mathrm{K,S}}(k, t \mid \boldsymbol{\Theta}^\star) = f_k(t \mid \boldsymbol{\Theta}^\star)\prod_{\substack{j=1\\j\neq k}}^{m} R_j(t \mid \boldsymbol{\Theta}^\star)\,, \tag{3.6}$$

where f_k *and* R_k *are respectively the probability density and reliability functions of the* k^{th} *component.*

Proof. See appendix A.1 for a detailed proof. However, a quick justification follows. The likelihood that component k is the cause of a system failure at time t is equivalent to the likelihood that component k fails at time t and the other components do not fail before time t. According to assumption 2, the random variables $\mathrm{T}_1, \ldots, \mathrm{T}_m$ are mutually independent, therefore the likelihood is the product of the probability density function $f_k(t \mid \boldsymbol{\Theta}_k^\star)$ and the reliability functions $R_j(t \mid \boldsymbol{\Theta}_j^\star)$ for $j \in \mathsf{K} \setminus \{k\}$. $\square$

By definition 2.6, if a system failure occurs at time t, then each component has a particular probability of being the cause as given by the following theorem.

Theorem 3.5. *The conditional probability mass function $p_{\mathrm{K} \mid \mathrm{S}}(k \mid t, \boldsymbol{\Theta}^\star)$ maps each component $k \in \{1, \ldots, m\}$ to the probability that component k is the cause of the system failure at time t and is given by*

$$p_{\mathrm{K} \mid \mathrm{S}}\left(k \mid t, \boldsymbol{\Theta}^\star\right) = \frac{h_k\left(t \mid \boldsymbol{\Theta}_k^\star\right)}{h_{\mathrm{S}}\left(t \mid \boldsymbol{\Theta}^\star\right)}\,, \tag{3.7}$$

where h_j and h_{S} are respectively the failure rate functions of the j^{th} component and the system.

Proof. By the axioms of probability,

$$p_{\mathrm{K} \mid \mathrm{S}}(k, \mid t, \boldsymbol{\Theta}^\star) = \frac{f_{\mathrm{K},\mathrm{S}}(k, t \mid \boldsymbol{\Theta}^\star)}{f_{\mathrm{S}}(t \mid \boldsymbol{\Theta}^\star)}\,, \tag{a}$$

thus we wish to show that

$$\frac{h_k(t \mid \boldsymbol{\Theta}_k^\star)}{h_{\mathrm{S}}(t \mid \boldsymbol{\Theta}^\star)} = \frac{f_{\mathrm{K},\mathrm{S}}(k, t \mid \boldsymbol{\Theta}^\star)}{f_{\mathrm{S}}(t \mid \boldsymbol{\Theta}^\star)}\,. \tag{b}$$

By theorem 3.3, the system's failure rate function is given by

$$h_{\mathrm{S}}(t \mid \boldsymbol{\Theta}^\star) = \sum_{j=1}^{m} h_j(t \mid \boldsymbol{\Theta}_j^\star)\,. \tag{3.5 revisited}$$

Performing this substitution results in

$$\frac{h_k(t \mid \boldsymbol{\Theta}_k^\star)}{h_{\mathrm{S}}(t \mid \boldsymbol{\Theta}^\star)} = \frac{h_k(t \mid \boldsymbol{\Theta}_k^\star)}{\sum_{j=1}^{m} h_j(t \mid \boldsymbol{\Theta}_j^\star)}\,. \tag{c}$$

By definition 3.2, the failure rate function of the p^{th} component is given by

$$h_p(t \mid \boldsymbol{\Theta}_p^\star) = \frac{f_p(t \mid \boldsymbol{\Theta}_p^\star)}{R_p(t \mid \boldsymbol{\Theta}_p^\star)}\,. \tag{d}$$

Performing this substitution results in

$$\frac{h_k(t \mid \boldsymbol{\Theta}_k^\star)}{h_{\mathrm{S}}(t \mid \boldsymbol{\Theta}^\star)} = \frac{f_k(t \mid \boldsymbol{\Theta}_k^\star)}{R_k(t \mid \boldsymbol{\Theta}_k^\star)} \cdot \left[\underbrace{\frac{f_1(t \mid \boldsymbol{\Theta}_1^\star)}{R_1(t \mid \boldsymbol{\Theta}_1^\star)} + \cdots + \frac{f_m(t \mid \boldsymbol{\Theta}_m^\star)}{R_m(t \mid \boldsymbol{\Theta}_m^\star)}}_{A}\right]^{-1}\,. \tag{e}$$

To combine the fractions labeled A in the above equation, we make their respective denominators the same, resulting in

$$\frac{h_k(t \mid \boldsymbol{\Theta}^\star_k)}{h_{\mathrm{S}}(t \mid \boldsymbol{\Theta}^\star)} = \underbrace{\frac{f_k(t \mid \boldsymbol{\Theta}^\star_k)}{R_k(t \mid \boldsymbol{\Theta}^\star_k)}}_{B} \cdot \frac{\overbrace{\prod_{j=1}^{m} R_j(t \mid \boldsymbol{\Theta}^\star)}^{C}}{\sum_{j=1}^{m} \left[\prod_{\substack{p=1\\p\neq j}}^{m} R_p(t \mid \boldsymbol{\Theta}^\star_p)\right] f_j(t \mid \boldsymbol{\Theta}^\star_j)}\,. \tag{f}$$

Dividing the part of the above equation labeled C in the numerator by the part labeled B in the denominator results in

$$\frac{h_k(t \mid \boldsymbol{\Theta}^\star_k)}{h_{\mathrm{S}}(t \mid \boldsymbol{\Theta}^\star)} = \frac{f_k(t \mid \boldsymbol{\Theta}^\star_k) \prod_{\substack{j=1\\j\neq k}}^{m} R_j(t \mid \boldsymbol{\Theta}^\star_j)}{\sum_{j=1}^{m} f_j(t \mid \boldsymbol{\Theta}^\star_j) \prod_{\substack{p=1\\p\neq j}}^{m} R_p(t \mid \boldsymbol{\Theta}^\star_p)}\,. \tag{g}$$

By eq. (3.3), the denominator in the above equation is the density $f_{\mathrm{S}}(t \mid \boldsymbol{\Theta}^\star)$ and, by eq. (3.6), the numerator is the joint density $f_{\mathrm{K,S}}(k, t \mid \boldsymbol{\Theta}^\star)$. Performing these substitutions results in

$$\frac{h_k(t \mid \boldsymbol{\Theta}^\star_k)}{h_{\mathrm{S}}(t \mid \boldsymbol{\Theta}^\star)} = \frac{f_{\mathrm{K,S}}(k, t \mid \boldsymbol{\Theta}^\star)}{f_{\mathrm{S}}(t \mid \boldsymbol{\Theta}^\star)}\,. \tag{h}$$

□

Corollary 3.5.1. *The joint probability density function of random variable* K *and random system lifetime* S *is given by*

$$f_{\mathrm{K,S}}\left(k, t \mid \boldsymbol{\Theta}^\star\right) = h_k\left(t \mid \boldsymbol{\Theta}^\star_k\right) R_{\mathrm{S}}\left(t \mid \boldsymbol{\Theta}^\star\right)\,. \tag{3.8}$$

The proof of eq. (3.8) immediately follows from eqs. (3.6) and (3.7).

3.3 Masked system failure times

According to definition 2.7, the random set $\mathcal{C}$ is a subset of components that are the top $\mathrm{W} = |\mathcal{C}|$ most probable causes of system failure. By assumption 4, the distribution of W is not dependent on the true parameter index $\boldsymbol{\Theta}^\star$.

Definition 3.3. *The probability that* $\mathrm{W} = |\mathcal{C}| = w$ *is given by the marginal probability mass function* $p_{\mathrm{W}}\left(w\right)$*.*

By assumption 4, the random lifetime S and the cardinality of the random candidate set $\mathrm{W} = |\mathcal{C}|$ are independent, thus by the axioms of probability the joint density of $\mathcal{C}$, S, and W is given by

$$f_{\mathcal{C},\mathrm{S},\mathrm{W}}(\mathsf{C}, t, w \mid \boldsymbol{\Theta}^\star) = p_{\mathcal{C} \mid \mathrm{S},\mathrm{W}}(\mathsf{C} \mid t, w, \boldsymbol{\Theta}^\star) p_{\mathrm{W}}(w) f_{\mathrm{S}}(t \mid \boldsymbol{\Theta}^\star)\,. \tag{3.9}$$

If it is given that $\mathrm{W} = w$, by the axioms of probability

$$f_{\mathcal{C},\mathrm{S} \mid \mathrm{W}}(\mathsf{C}, t \mid w, \boldsymbol{\Theta}^\star) = p_{\mathcal{C} \mid \mathrm{S},\mathrm{W}}(\mathsf{C} \mid t, w, \boldsymbol{\Theta}^\star) f_{\mathrm{S}}(t \mid \boldsymbol{\Theta}^\star)\,. \tag{3.10}$$

The indicator function is needed for subsequent material and is given by the following definition.

Definition 3.4 (Indicator function)**.** *Given a subspace* $\mathbf{E}$ *of a space* $\mathbf{A}$*, the indicator function* $\mathbb{1}_{\mathbf{E}} \colon \mathbf{A} \mapsto \{\mathbf{0}, \mathbf{1}\}$ *is equal to* 1 *if* $x \in \mathbf{E}$ *and* 0 *otherwise.*

Definition 3.5. *The conditional random variable $\mathcal{C} \mid \mathrm{W} = w$ has a sample space given by*

$$\mathcal{W}_w \equiv \{\mathsf{C} \in \mathcal{P}(\mathsf{K}) \mid |\mathsf{C}| = w\} \ , \tag{3.11}$$

the set of all candidate sets of cardinality w.

The conditional probability density of $\mathcal{C}$ and S given $\mathrm{W} = w$ is given by the following theorem.

Theorem 3.6. *The joint conditional probability density of $\mathcal{C}$ and S given $\mathrm{W} = w$ is given by*

$$f_{\mathcal{C},\mathrm{S}\mid\mathrm{W}}(\mathsf{C}, t \mid w, \mathbf{\Theta}^\star) = \eta R_{\mathrm{S}}\left(t \mid \mathbf{\Theta}^\star\right) \sum_{k \in \mathsf{C}} h_k\left(t \mid \mathbf{\Theta}^\star\right) \mathbb{1}_{\mathcal{W}_{\mathrm{w}}}(\mathsf{C})\,. \tag{3.12}$$

where η is the normalizing constant that makes the joint probability density function integrate[1] *to unity over the sample space $\mathcal{W}_w \times (0, \infty)$.*

Proof. Given a candidate set $\mathsf{C} \in \mathcal{W}_w$, the event E we wish to compute the likelihood of is given by

$$E = \bigcup_{j \in \mathsf{C}} \left(\mathrm{K} = j \cap \mathrm{S} = t\right) \ . \tag{a}$$

We would like to compute the likelihood $L(E)$. The likelihood that $\mathrm{K} = j$ and $\mathrm{S} = t$ is given by the probability density function

$$f_{\mathrm{K},\mathrm{S}}\left(k, t \mid \mathbf{\Theta}^\star\right) = h_k\left(t \mid \mathbf{\Theta}^\star_k\right) R_{\mathrm{S}}\left(t \mid \mathbf{\Theta}^\star\right) \ . \tag{3.8 revisited}$$

Performing this substitution yields

$$L(E) = \bigcup_{j \in \mathsf{C}} h_j\left(t \mid \mathbf{\Theta}^\star_j\right) R_{\mathrm{S}}\left(t \mid \mathbf{\Theta}^\star\right) \ . \tag{b}$$

Since the events $\mathrm{K} = j, \mathrm{S} = t$ for $j \in \mathsf{C}$ are mutually exclusive, the likelihood is given by

$$L(E) = \sum_{j \in \mathsf{C}} h_j\left(t \mid \mathbf{\Theta}^\star_j\right) R_{\mathrm{S}}\left(t \mid \mathbf{\Theta}^\star\right) \ . \tag{c}$$

Since outcomes in the sample space $\mathcal{W}_w$ are not atomic with respect to the sample space of $\mathrm{K} \in \{1, , \ldots , m\}$, we multiply by a normalizing constant η to make it a proper joint probability density function. □

The normalizing constant η indicated in eq. (3.12) is given by the following theorem.

Theorem 3.7. *The normalizing constant η that makes the joint probability density $f_{\mathcal{C},\mathrm{S}\mid\mathrm{W}}(\mathsf{C}, t \mid w, \mathbf{\Theta}^\star)$ integrate to unity over the sample space $\mathcal{W}_w \times (0, \infty)$ is given by*

$$\eta = \frac{1}{\binom{m-1}{w-1}}\,. \tag{3.13}$$

Proof. Integrating over the sample space $\mathcal{W}_w \times (0, \infty)$ must be equal to unity. That is,

$$\eta \sum_{\mathsf{C} \in \mathcal{W}_w} \int_0^\infty f_{\mathcal{C},\mathrm{S}\mid\mathrm{W}}(\mathsf{C}, t \mid \mathbf{\Theta}^\star)\, \mathrm{d}t = 1\,. \tag{a}$$

[1]Integration refers to both summations and integrations.

Substituting the joint probability density with the definition given by eq. (3.12) results in

$$\eta \sum_{\mathsf{C} \in \mathcal{W}_w} \int_0^\infty \left[\sum_{j \in \mathsf{C}} f_{\mathrm{K,S}}(j, t \mid \mathbf{\Theta}^\star) \right] \mathrm{d}t = 1\,. \tag{b}$$

We can switch the summation over $\mathcal{W}_w$ and the integration over t without changing the equation, resulting in

$$\eta \sum_{\mathsf{C} \in \mathcal{W}_w} \sum_{j \in \mathsf{C}} \left[\int_0^\infty f_{\mathrm{K,S}}(j, t \mid \mathbf{\Theta}^\star) \mathrm{d}t \right] = 1\,. \tag{c}$$

By the axioms of probability, the inner integral is the marginal probability mass function of K. Performing this substitution results in

$$\eta \sum_{\mathsf{C} \in \mathcal{W}_w} \sum_{j \in \mathsf{C}} p_{\mathrm{K}}(j \mid \mathbf{\Theta}^\star) = 1\,. \tag{d}$$

When constructing a candidate set of cardinality w, suppose we restrict our attention to only those sets in which component r is present. Then, we have $w-1$ additional components to choose from of the $m-1$ remaining. There are $\binom{m-1}{w-1}$ such choices possible, thus component r is present in $\binom{m-1}{w-1}$ of the sets. Consequently, each component has $\binom{m-1}{w-1}$ occurrences. Performing this substitution results in

$$\eta \binom{m-1}{w-1} \sum_{j=1}^{m} p_{\mathrm{K}}(j \mid \mathbf{\Theta}^\star) = 1\,. \tag{e}$$

Observe that $\sum_{j=1}^{m} p_{\mathrm{K}}(j \mid \mathbf{\Theta}^\star)$ must be equal to unity since we are summing over the entire sample space of K. Performing this substitution and solving for η yields

$$\eta = \frac{1}{\binom{m-1}{w-1}}\,. \tag{f}$$

□

Corollary 3.7.1. *The conditional probability mass function $p_{\mathcal{C} \mid \mathrm{S,W}}(\mathsf{C} \mid t, w, \mathbf{\Theta}^\star)$ maps each candidate set C to the probability that the given candidate set contains the component responsible for the system failure at time t. By the axioms of probability,*

$$p_{\mathcal{C} \mid \mathrm{S,W}}(\mathsf{C} \mid t, w, \mathbf{\Theta}^\star) = \frac{\sum_{j \in \mathsf{C}} h_j \left(t \mid \mathbf{\Theta}_j^\star\right)}{h_{\mathrm{S}}\left(t \mid \mathbf{\Theta}^\star\right)} \mathbb{1}_{\mathcal{W}_{\mathrm{w}}}(\mathsf{C})\,, \tag{3.14}$$

where $h_{\mathrm{S}}(\,\cdot \mid \mathbf{\Theta}^\star)$ and $h_j(\,\cdot \mid \mathbf{\Theta}_j^\star)$ are the failure rate functions of the system and the j^{th} component respectively.

Proof. By the axioms of probability,

$$p_{\mathcal{C} \mid \mathrm{S,W}}(\mathsf{C} \mid t, w, \mathbf{\Theta}^\star) = \frac{f_{\mathcal{C},\mathrm{S} \mid \mathrm{W}}(\mathsf{C}, t \mid w, \mathbf{\Theta}^\star)}{f_{\mathrm{S}}(t \mid \mathbf{\Theta}^\star)}\,, \tag{a}$$

where $f_{\mathcal{C},\mathrm{S} \mid \mathrm{W}}(\,\cdot \mid \mathbf{\Theta}^\star)$ is the joint probability density function given by theorem 3.6. Making this substitution immediately yields the result. □

Definition 3.6. *A random sample of n masked system failure times in which it is given that each system failure has w candidates, i.e., $\mathrm{S}_i, \mathcal{C}_i \,|\, \mathrm{W} = w$ for $i = 1, \ldots, n$, is denoted by $\mathbf{M}(w, n)$. That is, if given a random sample of masked system failure times, the conditional distribution $\mathbf{M}(w, n)$ is the subset of the random sample with w candidates.*

The likelihood of observing a particular realization, $\mathbf{M}(w, n) = \mathbf{m_n}$, is given by the following definition.

Theorem 3.8. *A random sample of n masked system failure times conditioned on candidate sets of cardinality w has a joint probability density given by*

$$f_{\mathbf{M}(w,n)}\left(\mathbf{m_n} \,|\, w, \mathbf{\Theta}^\star\right) = \prod_{i=1}^{n} f_{\mathrm{S}}(t_i \,|\, \mathbf{\Theta}^\star) \cdot p_{\mathcal{C}\,|\,\mathrm{S},\mathrm{W}}\left(\mathsf{C}_i \,|\, t_i, w, \mathbf{\Theta}^\star\right) \,, \tag{3.15}$$

where $f_{\mathrm{S}}(\cdot \,|\, \mathbf{\Theta}^\star)$ is the marginal density given by theorem 3.2 and $p_{\mathcal{C}\,|\,\mathrm{S},\mathrm{W}}(\cdot \,|\, t, w, , \mathbf{\Theta}^\star)$ is the conditional probability mass given by corollary 3.7.1.

The reason why a candidate set of cardinality w is generated is outside the scope of the statistical model and where necessary, the relative frequency in a sample is used.

The generative model of $\mathbf{M}(w, n)$, which directly follows from eq. (3.15), is given by algorithm 1.

Algorithm 1: Generative model of a masked system failure time with w candidates.

Result: a realization of a masked system failure time with w candidates.
Input:
$\mathbf{\Theta}^\star$, the true parameter value.
w, the cardinality of the candidate sets.
Output:
$\mathbf{m}$, a realization of $\mathcal{C}, \mathrm{S} \,|\, \mathrm{W} = w$.

Model `GenerateMSFT(`$w, \mathbf{\Theta}^\star$`)`
  draw a system lifetime $t \sim f_{\mathrm{S}}(\,\cdot \,|\, \mathbf{\Theta})$
  draw a candidate set $\mathsf{C} \sim p_{\mathcal{C}\,|\,\mathrm{S},\mathrm{W}}(\,\cdot \,|\, t, w, \mathbf{\Theta})$
  $\mathbf{m} \leftarrow (t, \mathsf{C})$
  return $\mathbf{m}$

4 Likelihood and Fisher information

In what follows, we frequently prefer to work with vectors rather than matrices.

Definition 4.1. *Let $\operatorname{vec}\colon \mathbb{R}^{m \times q} \mapsto \mathbb{R}^{m \cdot q}$ denote the linear transformation of a matrix of m rows and q columns into a column vector of $m \cdot q$ elements, where consecutive matrix rows are placed side by side.*

Notation. A matrix parameter index $\mathbf{\Theta}$ has a vector representation denoted by

$$\boldsymbol{\theta} \equiv \operatorname{vec}\left(\mathbf{\Theta}\right) \,, \tag{4.1}$$

thus the true matix parameter index $\mathbf{\Theta}^\star$ has a vector representation denoted by $\boldsymbol{\theta}^\star$. We use matrix and vector representations interchangeably.

The subsequent material makes a few assumptions, denoted *regularity conditions*, given by the following.

Assumption 8. *The following regularity conditions hold for the $\mathcal{F}_{\boldsymbol{\Theta}}$-family of the series system:*

1. *$\boldsymbol{\theta}^\star$ is interior to the parameter support Ω.*
2. *The log-likelihood ℓ is continuous, thrice differentiable, and bounded.*
3. *The true parameter value $\boldsymbol{\theta}^\star$ is identified, i.e.,*

$$\boldsymbol{\theta}^\star = \arg\max_{\boldsymbol{\theta} \in \Omega} \mathbb{E}_{\boldsymbol{\theta}^\star}\Big(\ln f_{\mathcal{C},\mathrm{S}\,|\,\mathrm{W}}\left(\mathcal{C}, \mathrm{S} \,|\, w, \boldsymbol{\theta}\right)\Big)\,. \tag{4.2}$$

By theorem 3.8, the probability density that $\mathbf{M}(w,n) = \mathbf{m_n}$ is given by

$$f_{\mathbf{M}(w,n)}(\mathbf{m_n} \,|\, w, \boldsymbol{\theta}^\star) = \prod_{i=1}^{n} f_{\mathcal{C},\mathrm{S}\,|\,\mathrm{W}}(\mathsf{C}_i, t_i \,|\, w, \boldsymbol{\theta}^\star)\,. \tag{3.15 revisited}$$

If we fix $\mathbf{m_n}$ and allow the parameter $\boldsymbol{\theta}$ to change, we have the likelihood function.

Definition 4.2. *The likelihood function represents the* likelihood *of parameter index $\boldsymbol{\theta}$ conditioned on $\mathbf{M}(w,n) = \mathbf{m_n}$ is given by*

$$\mathcal{L}(\boldsymbol{\theta} \,|\, w, \mathbf{m_n}) = \prod_{i=1}^{n} f_{\mathcal{C},\mathrm{S}\,|\,\mathrm{W}}(\mathsf{C}_i, t_i \,|\, w, \boldsymbol{\theta})\,. \tag{4.3}$$

The likelihood is a measure of the extent to which the given sample provides support for particular values of a parameter in $\mathcal{F}_{\boldsymbol{\Theta}}$.

The surface of the likelihood function captures all the information there is about $\boldsymbol{\theta}^\star$ in a sample. Any statistic that is equivalent to the likelihood function is a *sufficient* statistic. The likelihood function is a minimally sufficient statistic.

The log-likelihood plays a more central role than the likelihood for reasons that will be made clear later on. The log-likelihood function is given by the following theorem.

Definition 4.3. *The log-likelihood with respect to $\boldsymbol{\theta}$ given a particular $\mathbf{M}(w,n) = \mathbf{m_n}$ is given by*

$$\ell(\boldsymbol{\theta} \,|\, w, \mathbf{m_n}) = \sum_{i=1}^{n} \ln f_{\mathcal{C},\mathrm{S}\,|\,\mathrm{W}}(\mathsf{C}_i, t_i \,|\, w, \boldsymbol{\theta})\,. \tag{4.4}$$

In subsequent material, we need the gradient operator as given by the following definition.

Definition 4.4 (Gradient operator)**.** *The gradient of a function* $\mathrm{g}\colon \mathbb{R}^q \mapsto \mathbb{R}$ *with respect to column vector* $\mathbf{x} \in \mathbb{R}^q$ *is given by*

$$\nabla_{\mathbf{x}}\, \mathrm{g}\left(\mathbf{x}\right) = \left(\frac{\partial\, \mathrm{g}(\mathbf{x})}{\partial x_1}, \ldots, \frac{\partial\, \mathrm{g}(\mathbf{x})}{\partial x_q}\right)^{\mathsf{T}}\,. \tag{4.5}$$

The score function is given by the following definition.

Definition 4.5 (Score function)**.** *The gradient of the log-likelihood function with respect to $\boldsymbol{\theta}$ is the Fisher's score function is denoted by* $\mathrm{s}\colon \mathbb{R}^{m \cdot q} \mapsto \mathbb{R}^{m \cdot q}$*. That is,*

$$\mathrm{s}\left(\boldsymbol{\theta} \,|\, w, \mathbf{m_n}\right) = \nabla_{\boldsymbol{\theta}} \ell(\boldsymbol{\theta} \,|\, w, \mathbf{m_n})\,. \tag{4.6}$$

The variance-covariance is given by the following definition.

Definition 4.6 (Variance-covariance). *A random vector* $\mathbf{X} \in \mathbb{R}^p$ *has a variance-covariance given by*

$$\operatorname{Var}(\mathbf{X}) = \mathbb{E}\left[(\mathbf{X} - \mathbb{E}[\mathbf{X}])\,(\mathbf{X} - \mathbb{E}[\mathbf{X}])^{\intercal}\right] , \tag{4.7}$$

which is a p-by-p semi-positive definite matrix.

The score function when given a particular realization $\mathbf{m_n}$ is a statistic. When we consider it as a function of a random sample, it is a random vector. When evaluated at the true parameter index $\boldsymbol{\theta}^\star$, it has a variance-covariance given by the following definition.

Definition 4.7 (Fisher information matrix). *The score as a function of a random sample* $\mathbf{M}(w, n)$ *has a variance-covariance known as the Fisher information matrix, a* $(m \cdot q)$*-by-*$(m \cdot q)$ *matrix denoted by*

$$\boldsymbol{\mathcal{I}}_{\mathrm{n}}\left(\boldsymbol{\theta}^\star \,|\, w\right) = \operatorname{Var}_{\boldsymbol{\theta}^\star}\!\left[\mathrm{s}\left(\boldsymbol{\theta}^\star \,|\, w, \mathbf{M}(w,n)\right)\right] , \tag{4.8}$$

where the variance is taken with respect to the joint probability density function of $\mathcal{C}\,, \mathrm{S} \,|\, \mathrm{W} = w$ *indexed by* $\boldsymbol{\theta}^\star$*.*

The Hessian is given by the following definition.

Definition 4.8 (Hessian operator). *The Hessian of a function* $\mathrm{g} \colon \mathbb{R}^q \mapsto \mathbb{R}$ *with respect to* $\mathbf{x} \in \mathbb{R}^q$ *is a q-by-q matrix where the* (j,k)*-th element is given by*

$$\mathcal{H}\left(\mathrm{g}(\mathbf{x})\right)_{jk} = \frac{\partial^2\, \mathrm{g}(\mathbf{x})}{\partial x_j \partial x_k} . \tag{4.9}$$

Under the regularity conditions, the information matrix may be computed from the Hessian and is given by the following postulate.

Postulate 4.1. *The information matrix is given by the expectation of the negative of the Hessian of the log-likelihood function* ℓ *evaluated at* $\boldsymbol{\theta}^\star$*. That is,*

$$\boldsymbol{\mathcal{I}}_{\mathrm{n}}\left(\boldsymbol{\theta}^\star \,|\, w\right) = -\,\mathbb{E}_{\boldsymbol{\theta}^\star}\left[\mathcal{H}\left(\ell\left(\boldsymbol{\theta} \,|\, w, \mathbf{M}(w,n)\right)\right)\Big|_{\boldsymbol{\theta}=\boldsymbol{\theta}^\star}\right] , \tag{4.10}$$

where the expectation is taken with respect to $\boldsymbol{\theta}^\star$ *and the Hessian is taken with respect to* $\boldsymbol{\theta}$ *and then evaluated at* $\boldsymbol{\theta}^\star$*.*

For clarity, the (i,j)-th element of the information matrix is given by the following definition.

Definition 4.9. *The* (i,j)*-th element of* $\boldsymbol{\mathcal{I}}_{\mathrm{n}}(\,\cdot\,|\, w)$ *is given by*

$$\begin{aligned}\boldsymbol{\mathcal{I}}_{\mathrm{n}}(\boldsymbol{\theta}^\star \,|\, w)_{ij} = -\sum_{\mathsf{C} \in \mathcal{W}_w} \int_0^\infty \left[\frac{\partial^2}{\partial\theta_i \partial\theta_j} \ln f_{\mathcal{C},\mathrm{S}\,|\,\mathrm{W}}\left(\mathsf{C}_i, \mathrm{t_i} \,|\, \mathrm{w}, \boldsymbol{\theta}\right)\right]\Bigg|_{\boldsymbol{\theta}=\boldsymbol{\theta}^\star} \times \\ f_{\mathcal{C},\mathrm{S}\,|\,\mathrm{W}}\left(\mathsf{C}_i, \mathrm{t_i} \,|\, \mathrm{w}, \boldsymbol{\theta}^\star\right) \mathrm{d}t ,\end{aligned} \tag{4.11}$$

where $\mathcal{W}_w \times (0, \infty)$ *is the sample space.*

Notation. We denote the Fisher information matrix of $\boldsymbol{\theta}^\star$ given $\mathbf{M}(w, 1)$ by

$$\boldsymbol{\mathcal{I}}(\boldsymbol{\theta}^\star \,|\, w) \equiv \boldsymbol{\mathcal{I}}_{\mathbf{1}}(\boldsymbol{\theta}^\star \,|\, w) . \tag{4.12}$$

The information matrix is a key quantity in large sample theory; it quantifies the *expected* information (as a reduction of uncertainty) $\mathbf{M}(w, n)$ carries about $\boldsymbol{\theta}^\star$. For sufficiently large samples, the log-likelihood has a Taylor series approximation given by

$$\ell\left(\boldsymbol{\theta} \mid w\right) \approx -\frac{1}{2}\left(\boldsymbol{\theta} - \boldsymbol{\theta}^\star\right)^\intercal \boldsymbol{\mathcal{I}}_{\mathrm{n}}\left(\boldsymbol{\theta}^\star \mid w\right)\left(\boldsymbol{\theta} - \boldsymbol{\theta}^\star\right) , \tag{4.13}$$

which is maximized at $\boldsymbol{\theta}^\star$. Intuitively, the more "peaked" the log-likelihood is expected to be at $\boldsymbol{\theta}^\star$, the more information a random sample is expected to carry about $\boldsymbol{\theta}^\star$.

The information about $\boldsymbol{\theta}^\star$ of independent samples is *additive* as given by the following postulate.

Postulate 4.2. *Algebraically, the Fisher information about the true parameter index $\boldsymbol{\theta}^\star$ of independent samples is additive. That is,*

$$\boldsymbol{\mathcal{I}}_{\boldsymbol{n_1+n_2}}\left(\boldsymbol{\theta}^\star \mid \cdot\,\right) = \boldsymbol{\mathcal{I}}_{\boldsymbol{n_1}}(\boldsymbol{\theta}^\star \mid \cdot\,) + \boldsymbol{\mathcal{I}}_{\boldsymbol{n_2}}(\boldsymbol{\theta}^\star \mid \cdot\,) . \tag{4.14}$$

By postulate 4.2, the Fisher information of a random sample of n masked system failure times drawn from $\mathbf{M}(w, n)$ is given by

$$\boldsymbol{\mathcal{I}}_{\mathrm{n}}\left(\boldsymbol{\theta}^\star \mid w\right) = n\boldsymbol{\mathcal{I}}\left(\boldsymbol{\theta}^\star \mid w\right) . \tag{4.15}$$

A *masked system failure time* in which the cardinality of the candidate set is distributed according to W, as described by **??**, has a Fisher information matrix given by the following theorem.

Theorem 4.1. *The* Fisher information *matrix of true parameter index $\boldsymbol{\theta}^\star$ with respect to a realization of jointly distributed random system lifetime* $\mathrm{S} = t$ *and candidate set* $\mathcal{C} = \mathsf{C}$ *is given by*

$$\boldsymbol{\mathcal{I}}\left(\boldsymbol{\theta}^\star\right) = \sum_{w=1}^{m-1} \mathrm{p}_{\mathrm{W}}(w)\boldsymbol{\mathcal{I}}\left(\boldsymbol{\theta}^\star \mid w\right) , \tag{4.16}$$

where $\mathrm{p}_{\mathrm{W}}(w)$ *is the marginal distribution of* $\mathrm{W} = |\mathcal{C}|$.

Proof. Suppose we have a random sample of n masked system failure times. By **??**, the expected number of sample points that have w candidates is given by

$$n \cdot \mathrm{p}_{\mathrm{W}}(w) . \tag{a}$$

By the additive property of *Fisher information* given by postulate 4.2, the information matrix is given by

$$\boldsymbol{\mathcal{I}}_{\mathrm{n}}\left(\boldsymbol{\theta}^\star\right) = \sum_{w=1}^{m-1} n \cdot \mathrm{p}_{\mathrm{W}}(w)\boldsymbol{\mathcal{I}}\left(\boldsymbol{\theta}^\star \mid w\right) . \tag{b}$$

Setting n to 1 results in

$$\boldsymbol{\mathcal{I}}(\boldsymbol{\theta}^\star) = \sum_{w=1}^{m-1} \mathrm{p}_{\mathrm{W}}(w)\boldsymbol{\mathcal{I}}\left(\boldsymbol{\theta}^\star \mid w\right) . \tag{c}$$

□

Remark. A model is a simplification of reality. The model denoted by $\mathbf{M_n}$ is an approximation of some (unknown) underlying generative model, e.g., W and S may not truly be independent. Since the *expected* information matrix is an expectation with respect to the simplified model, the *expected* information is also a simplified approximation. The *observed* information matrix, given by

$$\mathcal{J}_{\mathrm{n}}(\hat{\boldsymbol{\theta}}_{\mathbf{n}} \mid \mathbf{m}_{\mathbf{n}}) = -\,\mathcal{H}\left(\ell\left(\boldsymbol{\theta} \mid \mathbf{m}_{\mathbf{n}}\right)\right)\Big|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}_{\mathbf{n}}}\,, \tag{4.17}$$

is a statistic of a real sample and therefore makes less of a commitment to the simplified model. Generally, the *observed* information matrix is preferable but since we know (by prescription) the underlying generative model,

$$\mathcal{J}_{\mathrm{n}}\left(\hat{\boldsymbol{\theta}}_{\mathbf{n}} \mid \mathbf{M}_{\mathbf{n}}\right) \xrightarrow{P} \mathcal{I}_{\mathrm{n}}(\boldsymbol{\theta}^{\star})\,, \tag{4.18}$$

and so we proceed with the *expected* information matrix.

5 Sampling distribution of parameter estimators

By definitions 2.4 and 2.5, the random system lifetime S has a probability density function that is a member of the $\mathcal{F}_{\boldsymbol{\Theta}}$-family with a true parameter index $\boldsymbol{\theta}^{\star}$. Let some estimator of $\boldsymbol{\theta}^{\star}$ be a function of the information in a random sample of n masked system failure times. The estimator is a function of a random sample, thus it has *sampling distribution*, a random vector denoted by $\mathbf{Y}_{\mathbf{n}}$. That is,

$$\mathbf{Y}_{\mathbf{n}} = \boldsymbol{\psi}\left(\mathbf{M}_{\mathbf{n}}\right)\,. \tag{5.1}$$

A particular realization of the estimator is denoted by $\bar{\boldsymbol{\theta}}_{\mathbf{n}}$, i.e., $\mathbf{Y}_{\mathbf{n}} = \bar{\boldsymbol{\theta}}_{\mathbf{n}}$. All else being equal, we prefer estimators which vary only *slightly* from sample to sample with a *central tendency* around $\boldsymbol{\theta}^{\star}$. That is, we prefer unbiased estimators in which each component has small variance.

The sampling distribution of $\bar{\boldsymbol{\theta}}_{\mathbf{n}}$ has a variance-covariance given by $\operatorname{Var}\left(\mathbf{Y}_{\mathbf{n}}\right)$ (see eq. (4.7)) and a bias given by the following definition.

Definition 5.1. *The bias of a point estimator* $\mathbf{Y}_{\mathbf{n}} = \bar{\boldsymbol{\theta}}_{\mathbf{n}}$ *is given by*

$$\operatorname{bias}\left(\mathbf{Y}_{\mathbf{n}}\right) = \mathbb{E}\left[\mathbf{Y}_{\mathbf{n}}\right] - \boldsymbol{\theta}^{\star}\,. \tag{5.2}$$

We are often faced with a trade-off between bias and variance. A measure of estimator accuracy that is a function of both the bias and the variance is the mean squared error as given by the following definition.

Definition 5.2 (MSE)**.** *The mean squared error of the sampling distribution of* $\bar{\boldsymbol{\theta}}_{\mathbf{n}}$ *is given by*

$$\operatorname{MSE}\left(\mathbf{Y}_{\mathbf{n}}\right) = \mathbb{E}\left[\left(\mathbf{Y}_{\mathbf{n}} - \boldsymbol{\theta}^{\star}\right)^{\intercal}\left(\mathbf{Y}_{\mathbf{n}} - \boldsymbol{\theta}^{\star}\right)\right] \tag{5.3}$$

An equivalent way to compute the mean squared error is given by the following postulate.

Postulate 5.1. *The mean squared error of the sampling distribution of* $\bar{\boldsymbol{\theta}}_{\mathbf{n}}$ *as given by definition 5.2 is equivalent to*

$$\operatorname{MSE}\left(\mathbf{Y}_{\mathbf{n}}\right) = \operatorname{tr}\left(\operatorname{Var}\left(\mathbf{Y}_{\mathbf{n}}\right)\right) + \operatorname{bias}^2\left(\mathbf{Y}_{\mathbf{n}}\right), \tag{5.4}$$

where $\operatorname{tr}\left(\boldsymbol{A}\right)$ *computes the sum of the diagonal elements of square matrix* $\boldsymbol{A}$*.*

Fisher information reduces the uncertainty about $\boldsymbol{\theta}^\star$. The *minimum-variance unbiased estimator* (UMVUE) has a lower-bound given by the inverse of the *Fisher information matrix*, denoted the *Cramér-Rao lower-bound*. The minimum variance obtainable from a random sample of n masked system failure times drawn from $\mathbf{M_n}$ is given by

$$\boldsymbol{CRLB_n} = \frac{1}{n}\boldsymbol{\mathcal{I}}^{-1}\left(\boldsymbol{\theta}^\star\right) . \tag{5.5}$$

By eq. (5.4), if $\bar{\boldsymbol{\theta}}_\mathbf{n}$ is an unbiased estimator of $\boldsymbol{\theta}^\star$, then the mean squared error is given by

$$\operatorname{MSE}\left(\mathbf{Y_n}\right) = \operatorname{tr}\left(\operatorname{Var}\left(\mathbf{Y_n}\right)\right) . \tag{5.6}$$

By eqs. (5.5) and (5.6), the mean squared error of any unbiased point estimator of $\boldsymbol{\theta}^\star$ in which the only *a priori* information is given by a random sample $\mathbf{M_n}$ has a lower-bound given by the trace of $\boldsymbol{CRLB_n}$,

$$\operatorname{MSE}\left(\mathrm{Y}\right) \geq \operatorname{tr}\left(\boldsymbol{CRLB_n}\right) . \tag{5.7}$$

5.1 Maximum likelihood estimator given w candidates

Suppose the only information about the true parameter index $\boldsymbol{\theta}^\star$ is given by a sample of n masked system failure times in which only those system failures with w candidates are observed. The maximum likelihood estimator based on this conditional sample is given in the following definition.

Definition 5.3. *A parameter index $\hat{\boldsymbol{\theta}}(n \mid w)$ that maximizes the* likelihood *of observing $\mathbf{M}(w, n) = \mathbf{m_n}$ (see definition 3.6) is a maximum likelihood estimator of $\boldsymbol{\theta}^\star$. That is,*

$$\hat{\boldsymbol{\theta}}(n \mid w) = \operatorname*{arg\,max}_{\boldsymbol{\theta} \in \boldsymbol{\Omega}} \mathcal{L}(\boldsymbol{\theta} \mid w, \mathbf{m_n}) , \tag{5.8}$$

where $\boldsymbol{\Omega}$ is the feasible parameter space of the parametric family.

As further justification of assumption 5, the maximum likelihood estimator maximizes the likelihood of generating the given $\mathbf{m_n}$ such that for each system failure time, a component in the corresponding candidate set is the cause.

The accuracy of the maximum likelihood estimator is a function of how indicative candidate sets are. For instance, we assume that the candidates are generated from the joint probability density of S and $\mathcal{C}$. If this is not a very accurate model (e.g., the inspector who constructed the candidate sets in the sample preferred to check a subset of the components for failure, and tended to neglect the others), the maximum likelihood estimator will be inconsistent since the model is not consistent with reality. Rather, the maximum likelihood estimator will generate an estimate that makes the provided sample the most likely to be generated by the assumed model, i.e., the component lifetimes will not be accurately modeled since they must be modeled in such a way as to make the biased sample more likely.

The logarithm is a monotonically increasing function, thus the parameter value $\hat{\boldsymbol{\theta}}(n \mid w)$ that maximizes $\mathcal{L}$ also maximizes the log-likelihood ℓ (see definition 4.3), i.e.,

$$\hat{\boldsymbol{\theta}}(n \mid w) = \operatorname*{arg\,max}_{\boldsymbol{\theta} \in \boldsymbol{\Omega}} \ell(\boldsymbol{\theta} \mid w, \mathbf{m_n}) . \tag{5.9}$$

According to Bickel [1], if the parameter support $\boldsymbol{\Omega}$ of the parametric family is open, the log-likelihood ℓ is differentiable with respect to $\boldsymbol{\theta}$, and $\hat{\boldsymbol{\theta}}(n \mid w)$ exists, then $\hat{\boldsymbol{\theta}}(n \mid w)$ must be a stationary point of ℓ as given by

$$\operatorname{s}\left(\hat{\boldsymbol{\theta}}(n \mid w) \mid w, \mathbf{m_n}\right) = \mathbf{0} , \tag{5.10}$$

where s is the score function given by eq. (4.6). In cases where there are no closed-form solutions to eq. (5.10), iterative methods may be used as described in appendix B.

An estimator of $\boldsymbol{\theta}^\star$ given a sample $\mathbf{M}(w,n) = \mathbf{m_n}$ is the maximum likelihood estimator $\hat{\boldsymbol{\theta}}(n \mid w)$ described in definition 5.3. Since $\hat{\boldsymbol{\theta}}(n \mid w)$ is a function of the random sample $\mathbf{M}(w,n)$, it has a *sampling distribution*.

Definition 5.4. *The* sampling distribution *of $\hat{\boldsymbol{\theta}}(n \mid w)$ is a random vector denoted by $\mathbf{Y}(n \mid w) \in \mathrm{R}^{m \cdot q}$.*

The generative model for the maximum likelihood estimator, `GenerateMLE`$(w \mid \cdot\,)$, is described by algorithm 2. This generative model depends on `GenerateMSFT`, the generative model for the *masked system failure time* as described by algorithm 1. Note that in algorithm 2, line 7 may be approximated with `FindMLE`, a function that numerically solves the stationary points of the maximum likelihood equations as described by algorithm 5.

Algorithm 2: Generative model of the maximum likelihood estimator conditioned on w candidates

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Result: a realization of a maximum likelihood estimate from the sampling distribution of $\hat{\boldsymbol{\theta}}(n \mid w)$.

Input:

$\boldsymbol{\theta}^\star$, the true parameter index.
w, the cardinality of the candidate set.
n, the number of masked system failure times.

Output:

$\hat{\boldsymbol{\theta}}(n \mid w)$, a realization of $\mathbf{Y}(n \mid w)$.

1 **Model** `GenerateMLE`$(n, w, \boldsymbol{\theta}^\star)$
2 $\quad \mathbf{m_n} \leftarrow \emptyset$
3 $\quad$ **for** $i \leftarrow 1$ **to** n **do**
4 $\quad\quad \mathbf{M} \leftarrow$ `GenerateMSFT` $(w, \boldsymbol{\theta}^\star)$
5 $\quad\quad \mathbf{m_n} \leftarrow \mathbf{m_n} \cup \{\mathbf{M}\}$
6 $\quad$ **end**
7 $\quad \hat{\boldsymbol{\theta}}(n \mid w) \leftarrow \arg\max_{\boldsymbol{\theta} \in \Omega} \ell\left(\boldsymbol{\theta} \mid w, \mathbf{m_n}\right)$
8 $\quad$ **return** $\hat{\boldsymbol{\theta}}(n \mid w)$

The asymptotic sampling distribution of $\hat{\boldsymbol{\theta}}(n \mid w)$ is a consistent estimator of $\boldsymbol{\theta}^\star$.

Postulate 5.2. *As $n \to \infty$, the sampling distribution of $\hat{\boldsymbol{\theta}}(n \mid w)$ converges in probability to $\boldsymbol{\theta}^\star$, written*

$$\mathbf{Y}(n \mid w) \xrightarrow{P} \boldsymbol{\theta}^\star\,, \tag{5.11}$$

since

$$\lim_{n \to \infty} \Pr\left[\mathrm{MSE}\left(\mathbf{Y}(n \mid w)\right) < \epsilon\right] = 1 \tag{5.12}$$

for every $\epsilon > 0$.

By eqs. (4.7) and (5.11), the variance-covariance of the asymptotic sampling distribution of $\hat{\boldsymbol{\theta}}(n \mid w)$ is given by

$$\operatorname{Var}\left[\mathbf{Y}(n \mid w)\right] = \mathbb{E}\left[\left(\mathbf{Y}(n \mid w) - \boldsymbol{\theta}^\star\right)\left(\mathbf{Y}(n \mid w) - \boldsymbol{\theta}^\star\right)^\intercal\right]\,. \tag{5.13}$$

Postulate 5.3. *The variance-covariance of the asymptotic sampling distribution of $\hat{\boldsymbol{\theta}}(n \mid w)$ obtains the* Cramér-Rao lower-bound *for point estimators that are strictly a function of n masked system failure times in which candidate sets are of cardinality w. We denote this variance-covariance by*

$$\Sigma_n(\boldsymbol{\theta}^\star \mid w) \equiv \frac{1}{n}\boldsymbol{\mathcal{I}}^{-1}(\boldsymbol{\theta}^\star \mid w)\ . \tag{5.14}$$

The asymptotic sampling distribution of $\hat{\boldsymbol{\theta}}(n \mid w)$ is normally distributed.

Postulate 5.4. *As $n \to \infty$, the sampling distribution of $\hat{\boldsymbol{\theta}}(n \mid w)$ converges in distribution to a multivariate normal distribution with a mean $\boldsymbol{\theta}^\star$ and a variance-covariance $\Sigma_\mathrm{n}(\boldsymbol{\theta}^\star \mid w)$, written*

$$\mathbf{Y}(n \mid w) \xrightarrow{d} \mathrm{MVN}\Big(\boldsymbol{\theta}^\star, \Sigma_n(\boldsymbol{\theta}^\star \mid w)\Big)\ . \tag{5.15}$$

The maximum likelihood estimator $\hat{\boldsymbol{\theta}}(n \mid w)$ is an asymptotically *efficient* estimator since it obtains the Cramér-Rao lower-bound as given by **??**. Thus, for a sufficiently large sample of masked system failure times, the sampling distribution of $\hat{\boldsymbol{\theta}}(n \mid w)$ varies only *slightly* from sample to sample with a *central tendency* around $\boldsymbol{\theta}^\star$.

By eqs. (5.5), (5.6) and (5.11) and **??**, the asymptotic sampling distribution of $\hat{\boldsymbol{\theta}}(n \mid w)$ has the minimum mean squared error of any unbiased estimator,

$$\mathrm{MSE}\left(\mathbf{Y}(n \mid w)\right) = \frac{1}{n}\,\mathrm{tr}\left(\boldsymbol{\mathcal{I}}^{-1}\Big(\boldsymbol{\theta}^\star \mid w\Big)\right) = \mathrm{tr}\left(\mathrm{CRLB}_{\mathrm{n\,w}}\right)\ . \tag{5.16}$$

Remark. Look up Slverty's theorem to justify the next paragraph.

To estimate the sampling distribution of $\hat{\boldsymbol{\theta}}(n \mid w)$, we may assume the sample size is sufficiently large such that the asymptotic distribution becomes a reasonable approximation. Since $\mathbf{Y}(n \mid w) \xrightarrow{P} \boldsymbol{\theta}^\star$ and $\mathbf{Y}(n \mid w) \xrightarrow{d} \mathrm{MVN}\left(\boldsymbol{\theta}^\star, \Sigma_\mathrm{n}\left(\boldsymbol{\theta}^\star \mid w\right)\right)$, it follows that

$$\mathbf{Y}(n \mid w) \xrightarrow{d} \mathrm{MVN}\left(\hat{\boldsymbol{\theta}}(n \mid w), \Sigma_n\left(\hat{\boldsymbol{\theta}}(n \mid w) \mid w\right)\right)\ . \tag{5.17}$$

Thus, we can approximate $\boldsymbol{\theta}^\star$ and $\boldsymbol{\mathcal{I}}\left(\boldsymbol{\theta}^\star \mid w\right)$ and obtain the following result. The proof of this theorem is beyond the scope of this paper.

Theorem 5.1. *For sufficiently large sample size n, the sampling distribution of $\hat{\boldsymbol{\theta}}(n \mid w)$ is approximately normally distributed with a mean $\hat{\boldsymbol{\theta}}(n \mid w)$ and a variance-covariance matrix $\Sigma_\mathrm{n}\left(\hat{\boldsymbol{\theta}}(n \mid w) \mid w\right)$, written*

$$\mathbf{Y}(n \mid w) \overset{approx.}{\sim} \mathrm{MVN}\left(\hat{\boldsymbol{\theta}}(n \mid w), \Sigma_n\left(\hat{\boldsymbol{\theta}}(n \mid w) \mid w\right)\right)\ . \tag{5.18}$$

5.2 Maximum likelihood estimator

Consider an i.i.d. sample of r asymptotically unbiased estimates of $\boldsymbol{\theta}^\star$ denoted by $\bar{\boldsymbol{\theta}}^{(i)}$ which have sampling distributions with variance-covariances given respectively by $\boldsymbol{\Sigma}^{(i)}$ for $i = 1, \ldots, r$. The maximum likelihood estimator of $\boldsymbol{\theta}^\star$ given these point estimates is the inverse-variance weighted mean and is given by

$$\hat{\boldsymbol{\theta}} = \left(\sum_{i=1}^{r} \boldsymbol{A_i}\right)^{-1} \left(\sum_{i=1}^{r} \boldsymbol{A_i}\bar{\boldsymbol{\theta}}^{(i)}\right)\ , \tag{5.19}$$

where $\boldsymbol{A_i}$ is the inverse of $\boldsymbol{\Sigma}^{(i)}$.

Suppose that the estimates are given by the maximum likelihood estimator described in **??**. The maximum likelihood estimator given these estimates has a sampling distribution given by the following definition.

Definition 5.5. *Let $\mathbf{M_n}$ be a random sample of n masked system failure times in which n_i realizations have w_i candidates for $i = 1, \ldots, r$. The maximum likelihood estimator given by*

$$\hat{\boldsymbol{\theta}}_{\mathbf{n}} = \hat{\boldsymbol{\theta}} = \left(\sum_{i=1}^{r} \boldsymbol{A_i}\right)^{-1} \left(\sum_{i=1}^{r} \boldsymbol{A_i}\bar{\boldsymbol{\theta}}^{(i)}\right), \tag{5.20}$$

has a sampling distribution given by

$$\mathbf{Y_n} = \left(\sum_{i=1}^{r} \boldsymbol{A_i}\right)^{-1} \left(\sum_{i=1}^{r} \boldsymbol{A_i}\mathbf{Y}^{(i)}(n_i \mid w_i)\right), \tag{5.21}$$

where $\mathbf{Y}^{(i)}(n_i \mid w_i)$ is the sampling distribution of $\hat{\boldsymbol{\theta}}(n_i \mid w_i)$ for $i = 1, \ldots, r$ and

$$\boldsymbol{A_i} = n_i \boldsymbol{\mathcal{I}}\left(\boldsymbol{\theta}^\star \mid w_i\right) \tag{5.22}$$

is the information matrix for $\boldsymbol{\theta}^\star$ with respect to $\mathbf{M}(w_i, n_i)$.

Theorem 5.2. *The random vector $\mathbf{Y_n}$ is an asymptotically unbiased estimator of $\boldsymbol{\theta}^\star$.*

Proof. The expectation of $\mathbf{Y_n}$ is given by

$$\begin{aligned}
\mathbb{E}\left[\mathbf{Y_n}\right] &= \mathbb{E}\left[\left(\sum_{i=1}^{r} \boldsymbol{A_i}\right)^{-1} \left(\sum_{i=1}^{r} \boldsymbol{A_i}\mathbf{Y}^{(i)}(n_i \mid w_i)\right)\right] && \text{(a)}\\
&= \left(\sum_{i=1}^{r} \boldsymbol{A_i}\right)^{-1} \left(\sum_{i=1}^{r} \boldsymbol{A_i}\,\mathbb{E}\left[\mathbf{Y}^{(i)}(n_i \mid w_i)\right]\right) && \text{(b)}\\
&= \left(\sum_{i=1}^{r} \boldsymbol{A_i}\right)^{-1} \left(\sum_{i=1}^{r} \boldsymbol{A_i}\right)\boldsymbol{\theta}^\star && \text{(c)}\\
&= \boldsymbol{\theta}^\star\,. && \text{(d)}
\end{aligned}$$

□

Theorem 5.3. *The variance-covariance of $\mathbf{Y_n}$ is given by*

$$\operatorname{Var}\left[\mathbf{Y_n}\right] = \left(\sum_{i=1}^{r} n_i \boldsymbol{\mathcal{I}}\left(\boldsymbol{\theta}^\star \mid w_i\right)\right)^{-1}. \tag{5.23}$$

Proof. Let

$$\boldsymbol{B} = \left(\sum_{i=1}^{r} \boldsymbol{A_i}\right)^{-1}. \tag{a}$$

The variance-covariance of $\mathbf{Y_n}$ is given by

$$\begin{aligned}
\operatorname{Var}\left[\mathbf{Y_n}\right] &= \operatorname{Var}\left[\boldsymbol{B}\left(\sum_{i=1}^{r}\boldsymbol{A_i}\mathbf{Y}^{(i)}(n_i \mid w_i)\right)\right] && \text{(b)}\\
&= \boldsymbol{B}\operatorname{Var}\left[\sum_{i=1}^{r}\boldsymbol{A_i}\mathbf{Y}^{(i)}(n_i \mid w_i)\right]\boldsymbol{B}^{\mathsf{T}} && \text{(c)}\\
&= \boldsymbol{B}\left(\sum_{i=1}^{r}\operatorname{Var}\left[\boldsymbol{A_i}\mathbf{Y}^{(i)}(n_i \mid w_i)\right]\right)\boldsymbol{B}^{\mathsf{T}} && \text{(d)}\\
&= \boldsymbol{B}\left(\sum_{i=1}^{r}\boldsymbol{A_i}\operatorname{Var}\left[\mathbf{Y}^{(i)}(n_i \mid w_i)\right]\boldsymbol{A_i}^{\mathsf{T}}\right)\boldsymbol{B}^{\mathsf{T}}. && \text{(e)}
\end{aligned}$$

By **??**, the variance-covariance of $\mathbf{Y}^{(i)}(n_i \mid w_i)$ is given by

$$\frac{1}{n_i}\boldsymbol{\mathcal{I}}^{-1}\left(\boldsymbol{\theta}^{\star} \mid w_i\right). \tag{f}$$

By eq. (5.22), this is equivalent to $\boldsymbol{A_i^{-1}}$. Performing this substitution results in

$$\begin{aligned}
\operatorname{Var}\left[\mathbf{Y_n}\right] &= \boldsymbol{B}\left(\sum_{i=1}^{r}\boldsymbol{A_i}\boldsymbol{A_i^{-1}}\boldsymbol{A_i}^{\mathsf{T}}\right)\boldsymbol{B}^{\mathsf{T}} && \text{(g)}\\
&= \boldsymbol{B}\left(\sum_{i=1}^{r}\boldsymbol{A_i}^{\mathsf{T}}\right)\boldsymbol{B}^{\mathsf{T}}. && \text{(h)}
\end{aligned}$$

The summation is equivalent to $\left(\boldsymbol{B}^{\mathsf{T}}\right)^{\boldsymbol{-1}}$. Performing this substitution results in

$$\begin{aligned}
\operatorname{Var}\left[\mathbf{Y_n}\right] &= \boldsymbol{B}\left(\boldsymbol{B}^{\mathsf{T}}\right)^{\boldsymbol{-1}}\boldsymbol{B}^{\mathsf{T}} && \text{(i)}\\
&= \boldsymbol{B} && \text{(j)}
\end{aligned}$$

By eq. (a), $\boldsymbol{B}$ is given by

$$\left(\sum_{i=1}^{r}\boldsymbol{A_i}\right)^{-1} \tag{k}$$

and by eq. (5.22), $\boldsymbol{A_i}$ is given by

$$n_i\boldsymbol{\mathcal{I}}\left(\boldsymbol{\theta}^{\star} \mid w_i\right). \tag{l}$$

Performing these substitution results in

$$\operatorname{Var}\left[\mathbf{Y_n}\right] = \left(\sum_{i=1}^{r}n_i\boldsymbol{\mathcal{I}}\left(\boldsymbol{\theta}^{\star} \mid w_i\right)\right)^{-1}. \tag{m}$$

□

The weighted estimator asymptotically achieves the Cramér-Rao lower-bound as given by eq. (5.5) thus it is the asymptotic UMVUE estimator of $\boldsymbol{\theta}^\star$ given a random sample of masked system failure times in which n_i realizations have w_i candidates for $i = 1, \ldots, r$.

A linear combination of multivariate normal distributions is a multivariate normal distribution, thus the asymptotic sampling distribution of $\hat{\boldsymbol{\theta}}_{\mathbf{n}}$ is normally distributed.

Postulate 5.5. *As $n \to \infty$, the sampling distribution of $\hat{\boldsymbol{\theta}}_{\mathbf{n}}$ converges in distribution to a multivariate normal with a mean $\boldsymbol{\theta}^\star$ and a variance-covariance given by eq. (5.23), written*

$$\mathbf{Y_n} \xrightarrow{d} \operatorname{MVN}\left(\boldsymbol{\theta}^\star, \left(\sum_{i=1}^{r} n_i \boldsymbol{\mathcal{I}}\left(\boldsymbol{\theta}^\star \mid w_i\right)\right)^{-1}\right) . \tag{5.24}$$

The generative model for $\mathbf{Y_n}$ is given by algorithm 4.

Algorithm 3: Generative model of maximum likelihood estimator

---

Input:
$\boldsymbol{\Theta}^\star$, the true parameter value of the series system.
Output:
$\hat{\boldsymbol{\theta}}_{\mathbf{n}}$, a realization of $\mathbf{Y_n}$.

```
Model GenerateMLE(Θ*)
    draw candidate cardinality w ~ p_W(·)
    θ̂_n ← GenerateMLE(Θ*, w)
    return θ̂_n
```

5.2.1 Bootstrap of sample covariance

Another estimator of the variance-covariance of $\mathbf{Y_n}$ is given by the sample covariance.

Definition 5.6. *Given a sample of r maximum likelihood estimates, $\hat{\boldsymbol{\theta}}_{\mathbf{n}}^{(\mathbf{1})}, \ldots, \hat{\boldsymbol{\theta}}_{\mathbf{n}}^{(\mathbf{r})}$, the maximum likelihood estimator of the variance-covariance of the sampling distribution of $\hat{\boldsymbol{\theta}}_{\mathbf{n}}$ is given by the sample covariance,*

$$\hat{\operatorname{Var}}\left[\mathbf{Y_n}\right] = \frac{1}{r} \sum_{i=1}^{r} \left(\hat{\boldsymbol{\theta}}_{\mathbf{n}}^{(\mathbf{i})} - \boldsymbol{\theta}^\star\right) \cdot \left(\hat{\boldsymbol{\theta}}_{\mathbf{n}}^{(\mathbf{i})} - \boldsymbol{\theta}^\star\right)^{\intercal} . \tag{5.25}$$

The sample covariance depends upon a random sample of maximum likelihood estimates, and thus the sample covariance is a random matrix. However, by the property that maximum likelihood estimators converge in probability to the true value,

$$\hat{\operatorname{Var}}\left[\mathbf{Y_n}\right] \xrightarrow{P} \operatorname{Var}\left[\mathbf{Y_n}\right] , \tag{5.26}$$

it follows that

$$\mathbf{Y_n} \xrightarrow{d} \operatorname{MVN}\left(\boldsymbol{\theta}^\star, \hat{\operatorname{Var}}\left[\mathbf{Y_n}\right]\right) . \tag{5.27}$$

If only one maximum likelihood estimate is realized, $\mathbf{Y_n} = \hat{\boldsymbol{\theta}}_{\mathbf{n}}$, then the sample covariance given by **??** cannot be computed. In the *parametric Bootstrap*, the sample covariance is approximated by using $\hat{\boldsymbol{\theta}}_{\mathbf{n}}$ as an estimate of $\boldsymbol{\theta}^\star$ in the generative model described by **??**, i.e.,

$$\hat{\boldsymbol{\theta}}_{\mathbf{n}}^{(\mathbf{1})}, \ldots, \hat{\boldsymbol{\theta}}_{\mathbf{n}}^{(\mathbf{r})} \leftarrow \texttt{GenerateMLE}\left(\hat{\boldsymbol{\theta}}_{\mathbf{n}}\right) , \tag{5.28}$$

and the sample covariance of these approximate maximum likelihood estimates is an asymptotically unbiased estimator of $\mathrm{Var}\left[\mathbf{Y_n}\right]$.

Algorithm 4: Sample covariance of a bootstrapped sample of maximum likelihood estimates.

Input:
$\hat{\boldsymbol{\theta}}_\mathbf{n}$, an estimate of $\boldsymbol{\theta}^\star$.

Output:
sample covariance of a bootstrapped sample of maximum likelihood estimates.

Model *BoostrapCovariance*
  $(\hat{\boldsymbol{\theta}}_\mathbf{n})$
for $i = 1$ *to* r **do**
  $\hat{\boldsymbol{\theta}}_\mathbf{n}^{(\mathbf{i})} \leftarrow \texttt{GenerateMLE}(\hat{\boldsymbol{\theta}}_\mathbf{n})$
end
return *sample covariance of* $\hat{\boldsymbol{\theta}}_\mathbf{n}^{(\mathbf{1})}, \ldots, \hat{\boldsymbol{\theta}}_\mathbf{n}^{(\mathbf{r})}$

---

6 Sampling distribution of functions of parameters

Suppose we have a characteristic of interest that is a function of $\boldsymbol{\theta}^\star \in \mathbb{R}^{m \cdot q}$, denoted by $\mathbf{g}(\boldsymbol{\theta}^\star) \colon \mathbb{R}^{m \cdot q} \mapsto \mathbb{R}^p$. By the invariance property of maximum likelihood estimators, if $\hat{\boldsymbol{\theta}}_\mathbf{n}$ is the maximum likelihood estimator of $\boldsymbol{\theta}^\star$ then $\mathbf{g}(\hat{\boldsymbol{\theta}}_\mathbf{n})$ is the maximum likelihood estimator of $\mathbf{g}(\boldsymbol{\theta}^\star)$.

By postulate 5.4, $\mathbf{Y_n}$ is a random vector drawn from a multivariate normal distribution,

$$\mathbf{Y_n} \sim \mathrm{MVN}\left(\boldsymbol{\theta}^\star, \mathrm{Var}\left[\mathbf{Y_n}\right]\right) , \tag{5.24 revisited}$$

Therefore, $\mathbf{g}\left(\mathbf{Y_n}\right)$ is a random vector. Under the regularity conditions (see assumption 8), $\mathbf{g}\left(\mathbf{Y_n}\right)$ is asymptotically normally distributed with a mean given by $\mathbf{g}(\boldsymbol{\theta}^\star)$ and a variance-covariance given by $\mathrm{Var}\left[\mathbf{g}(\mathbf{Y_n})\right]$, written

$$\mathbf{g}\left(\mathbf{Y_n}\right) \xrightarrow{d} \mathrm{MVN}\left(\mathbf{g}\left(\boldsymbol{\theta}^\star\right), \mathrm{Var}\left[\mathbf{g}(\mathbf{Y_n})\right]\right) . \tag{6.1}$$

6.1 Confidence intervals

A property of multivariate normal distributions is that the marginal of any subset of the components is given by dropping the irrelevant components from the mean vector $\boldsymbol{\theta}^\star$ and the variance-covariance $\mathrm{Var}\left[\mathbf{Y_n}\right]$.

If we are interested in the asymptotic marginal distribution of the j^{th} component, we drop all of the other components, resulting in the univariate normal distribution given by

$$\left[\mathbf{Y_n}\right]_j \sim \mathcal{N}\left(\theta_j^\star, \frac{\sigma^2}{n}\right) , \tag{6.2}$$

where $\sigma^2 = \left[\boldsymbol{\mathcal{I}}^{-1}\left(\boldsymbol{\theta}^\star\right)\right]_{jj}$.

Asymptotically, the random variable $[\mathbf{Y_n}]_j$ realizes value less than constant a with probability given by

$$\Pr\left[[\mathbf{Y_n}]_j < a\right] = \Phi\left(\frac{a - \boldsymbol{\theta}^\star_j}{\sigma_j}\right) , \tag{6.3}$$

where Φ is the cumulative distribution function of the standard normal.

The smallest interval in which $[\mathbf{Y_n}]_j$ is realized with probability p is given by

$$\boldsymbol{\theta}^\star_j \pm \sigma_j \, \Phi^{-1}(p/2) , \tag{6.4}$$

where Φ^{-1} is the inverse cumulative distribution function of the standard normal.

By the invariance property of maximum likelihood estimators, a maximum likelihood estimator of eq. (6.4) is the confidence interval given by the following definition.

Definition 6.1. *An asymptotic $(1-\alpha)\cdot 100\%$ confidence interval for the j^{th} component of the true parameter index $\boldsymbol{\theta}^\star$ is given by*

$$\hat{\theta}_j \pm \hat{\sigma}_j \;\Phi^{-1}(1-\alpha/2) , \tag{6.5}$$

where $\hat{\theta}_j$ is the j^{th} component of $\hat{\boldsymbol{\theta}}_\mathbf{n}$ and $\hat{\sigma}_j$ is the j^{th} diagonal element of $\hat{\mathrm{Var}}\,[\mathbf{Y_n}]$.

Note that the confidence interval given by eq. (6.5) is a realization of a random interval since $\hat{\theta}_j$ is a realization of the normal distribution given by eq. (6.2) and $\hat{\sigma}_j$ is a function $\hat{\theta}_j$, where it is expected that $(1-\alpha/2)\cdot 100\%$ of the random intervals generated contain the true parameter index $\theta^\star_j$.

6.2 Hypothesis testing

The intervals discussed in section 6.1 ignore correlations between the components of $\mathbf{Y_n}$. Asymptotically, $\mathbf{Y_n}$ has a probability p of occurring within the hyper-ellipsoid (the smallest hyper-volume with probability p) centered around $\boldsymbol{\theta}^\star$ given by

$$\left(\mathbf{Y_n} - \boldsymbol{\theta}^\star\right)^\intercal \boldsymbol{\mathcal{I}}_\mathrm{n}\left(\boldsymbol{\theta}^\star\right)\left(\mathbf{Y_n} - \boldsymbol{\theta}^\star\right) \;\le\; \chi^2_{m\cdot q}(p) , \tag{6.6}$$

where $\chi^2_{m\cdot q}(p)$ is the p quantile of the chi-square distribution with $m\cdot q$ degrees of freedom.

By the invariance property of maximum likelihood estimators, a maximum likelihood estimator of eq. (6.6) is given by the confidence region.

Definition 6.2. *Suppose we have a maximum likelihood estimate $\hat{\boldsymbol{\theta}}_\mathbf{n}$ and hypothesize that the true parameter index is given by $\boldsymbol{\theta}$. At significance level α we fail to reject the null hypothesis $H_0\colon \boldsymbol{\theta}^\star = \boldsymbol{\theta}$ versus the alternative hypothesis $H_a\colon \boldsymbol{\theta}^\star \ne \boldsymbol{\theta}$ if*

$$\left(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}_\mathbf{n}\right)^\intercal \boldsymbol{\mathcal{I}}_\mathrm{n}\left(\hat{\boldsymbol{\theta}}_\mathbf{n}\right)\left(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}_\mathbf{n}\right) \;\le\; \chi^2_{m\cdot q}\left(1-\alpha\right) , \tag{6.7}$$

where $\chi^2_{m\cdot q}(1-\alpha)$ is the $(1-\alpha)$ quantile of the chi-square distribution with $m\cdot q$ degrees of freedom.

Applying eq. (6.7) to the marginal of $\left[\mathbf{Y}(n\,|\,w)\right]_j$ generates the confidence interval described in **??**.

6.3 Monte-carlo simulation

The generative model of the sampling distribution of $\mathbf{Y_n}$, as described in algorithm 4, is asymptotically equivalent to generating samples from the multivariate normal distribution given by **??**. However, generating samples from the multivariate normal is much less computationally demanding than the non-asymptotic model.

The generative model may be used to generate samples of any statistic that is a function of the true parameter index $\boldsymbol{\theta}^\star$. A sample drawn from $\mathbf{g}(\mathbf{Y}(n \mid w))$ may be generated by drawing r maximum likelihood estimates from the sampling distribution

$$\hat{\boldsymbol{\theta}}_{\mathbf{n}}^{(\mathbf{1})}, \ldots, \hat{\boldsymbol{\theta}}_{\mathbf{n}}^{(\mathbf{r})}, \tag{6.8}$$

and applying $\mathbf{g}$ to each, resulting in the sample

$$\mathbf{g}\left(\hat{\boldsymbol{\theta}}_{\mathbf{n}}^{(\mathbf{1})}\right), \ldots, \mathbf{g}\left(\hat{\boldsymbol{\theta}}_{\mathbf{n}}^{(\mathbf{r})}\right). \tag{6.9}$$

An estimate of the variance-covariance $\operatorname{Var}\left[\mathbf{g}(\mathbf{Y_n}\right]$ is given by the sample covariance

$$\begin{aligned}\hat{\operatorname{Var}}\left[\mathbf{g}\left(\mathbf{Y_n}\right)\right] = \\ \frac{1}{r}\sum_{i=1}^{r}\left(\mathbf{g}\left(\hat{\boldsymbol{\theta}}_{\mathbf{n}}^{(\mathbf{i})}\right) - \mathbf{g}(\hat{\boldsymbol{\theta}}_{\mathbf{n}})\right)\left(\mathbf{g}\left(\hat{\boldsymbol{\theta}}_{\mathbf{n}}^{(\mathbf{i})}\right) - \mathbf{g}(\hat{\boldsymbol{\theta}}_{\mathbf{n}})\right)^{\intercal}.\end{aligned} \tag{6.10}$$

Thus,

$$\mathbf{g}\left(\mathbf{Y_n}\right) \sim \operatorname{MVN}\left(\mathbf{g}\left(\hat{\boldsymbol{\theta}}_{\mathbf{n}}\right), \hat{\operatorname{Var}}\left[\mathbf{Y_n}\right]\right). \tag{6.11}$$

In what follows, we explore a couple of characteristics of interest about the series system.

6.3.1 Mean time to failure of components

The *mean time to failure* of component j is given by

$$\mathrm{g}(\boldsymbol{\theta}^\star) = \mathbb{E}_{\boldsymbol{\theta}^\star}[\mathrm{T}_j]. \tag{6.12}$$

Suppose $\mathrm{g}(\boldsymbol{\theta}^\star) \in \mathbb{R}^{m \cdot q}$, then the variance-covariance of

$$\mathcal{J}\left(\mathrm{g}(\boldsymbol{\theta}^\star)\right)\boldsymbol{\mathcal{I}}_{\mathrm{n}}^{-1}\left(\boldsymbol{\theta}^\star\right)\mathcal{J}\left(\mathrm{g}(\boldsymbol{\theta}^\star)\right)^{\intercal}, \tag{6.13}$$

where $\mathcal{J}\left(\,\cdot\,\right) \in \mathbb{R}^{?}$ is the Jacobian as given by **??**.

The maximum likelihood estimator of $\boldsymbol{\theta}^\star$ is given by $\mathrm{g}(\hat{\boldsymbol{\theta}}_{\mathbf{n}})$ and is approximately normally distributed,

$$\mathrm{g}(\mathbf{Y_n}) \xrightarrow{d} \mathcal{N}\left(\mathrm{g}\left(\hat{\boldsymbol{\theta}}_{\mathbf{n}}\right), \hat{\sigma}^2\right). \tag{6.14}$$

where $\hat{\sigma}^2$ is the sample variance.

7 Expoentially distributed components

We consider series systems in which component lifetimes are exponentially distributed and compare them to the constrained maximum entropy distribution, which consists of identical exponentially distributed component lifetimes.

7.1 Parametric functions

Suppose the j^{th} component has an exponentially distributed lifetime and a failure rate $\lambda_j^\star$, denoted by $\mathrm{T}_j \sim \mathrm{EXP}(\lambda_j^\star)$. The random variable T_j has reliability, density, and failure rate functions given respectively by

$$R_j(t \mid \lambda_j^\star) = \exp(-\lambda_j^\star \cdot t)\,, \tag{7.1}$$

$$f_j(t \mid \lambda_j^\star) = \lambda_j^\star \exp(-\lambda_j^\star \cdot t)\,, \tag{7.2}$$

$$h_j(\cdot \mid \lambda_j^\star) = \lambda_j^\star\,, \tag{7.3}$$

where $t > 0$ and $\lambda_j^\star > 0$.

An m-out-of-m system with exponentially distributed lifetimes is also exponentially distributed as proved by the following theorem.

Theorem 7.1. *The random lifetime* S *of a series system composed of m components with exponentially distributed lifetimes is exponentially distributed with a failure rate that is the sum of the component failure rates with a true parameter index given by*

$$\boldsymbol{\lambda}^\star = \langle \lambda_1^\star, \ldots, \lambda_m^\star \rangle^\intercal \tag{7.4}$$

where $\lambda_j^\star$ is the failure rate of the j^{th} component. That is,

$$\mathrm{S} \sim \mathrm{EXP}\left(\sum_{j=1}^{m} \lambda_j^\star\right)\,, \tag{7.5}$$

Proof. By theorem 3.1,

$$R_{\mathrm{S}}(t \mid \boldsymbol{\lambda}^\star) = \prod_{j=1}^{m} R_j(t \mid \lambda_j^\star)\,. \tag{a}$$

Plugging in the component reliability functions as given by eq. (7.1) results in

$$R_{\mathrm{S}}(t \mid \boldsymbol{\lambda}^\star) = \prod_{j=1}^{m} \exp(-\lambda_j^\star \cdot t) = \exp\left(-\left[\sum_{j=1}^{m} \lambda_j^\star\right] t\right)\,. \tag{b}$$

This reliability function belongs to the family of exponential distributions with a true parameter index given by the sum of the component failure rates. □

By theorem 7.1, the system has a probability density, failure rate, and reliability functions given respectively by

$$f_{\mathrm{S}}(t \mid \boldsymbol{\lambda}^\star) = \left(\sum_{j=1}^{m} \lambda_j^\star\right) \exp\left(-\left[\sum_{j=1}^{m} \lambda_j{}^\star\right] t\right)\,, \tag{7.6}$$

$$h_{\mathrm{S}}(\cdot \mid \boldsymbol{\lambda}^\star) = \sum_{j=1}^{m} \lambda_j^\star\,, \tag{7.7}$$

$$R_{\mathrm{S}}\,(t \mid \boldsymbol{\lambda}^\star) = \exp\left(-\left[\sum_{j=1}^{m} \lambda_j^\star\right] t\right) \tag{7.8}$$

where $t > 0$ and $\lambda_j^\star > 0$.

The conditional probability that component k is the cause of a system failure at time t is given by

$$p_{\mathrm{K}\,|\,\mathrm{S}}(k \,|\, t, \boldsymbol{\lambda}^\star) = p_{\mathrm{K}}(k \,|\, \boldsymbol{\lambda}^\star) = \frac{\lambda_k^\star}{\sum_{p=1}^m \lambda_p^\star} \mathbb{1}_{\mathrm{k} \in \{1,\ldots,\mathrm{m}\}} \,. \tag{7.9}$$

Proof. By theorem 3.3,

$$p_{\mathrm{K}\,|\,\mathrm{S}}(k \,|\, t, \boldsymbol{\lambda}^\star) = \frac{h_k(t \,|\, \boldsymbol{\lambda}_k^\star)}{h_{\mathrm{S}}(t \,|\, \boldsymbol{\lambda}^\star)} \,. \tag{a}$$

Plug in the failure rate of component k and the failure rate of the system given by eqs. (7.3) and (7.7). ☐

Due to the constant failure rates of the components, K and S are independent. The joint density of K and S is given by

$$f_{\mathrm{K,S}}(k, t \,|\, \boldsymbol{\lambda}^\star) = \lambda_k^\star \exp\left(- \left[\sum_{j=1}^m {\lambda_j}^\star \right] t \right) \mathbb{1}_{\mathrm{t}>0} \, \mathbb{1}_{\mathsf{K} \times \mathbb{R}^+}(k, t) \tag{7.10}$$

Proof. By definition,

$$f_{\mathrm{K,S}}(k, t \,|\, \boldsymbol{\lambda}^\star) = p_{\mathrm{K}\,|\,\mathrm{S}}(k \,|\, t, \boldsymbol{\lambda}^\star) f_{\mathrm{S}}(t \,|\, \boldsymbol{\lambda}^\star) \,. \tag{a}$$

Plug in the conditional probability and the marginal probability given by eqs. (7.6) and (7.9). ☐

The conditional joint density of $\mathcal{C}$ and S given $\mathrm{W} = w$ is given by

$$f_{\mathcal{C},\mathrm{S}\,|\,\mathrm{W}}(\mathsf{C}, t \,|\, w, \boldsymbol{\lambda}^\star) = \frac{\mathbb{1}_{\mathrm{t}>0} \, \mathbb{1}_{|\mathsf{C}|=\mathrm{w}}}{\binom{m-1}{w-1}} \left(\sum_{j \in \mathsf{C}} \lambda_j \right) \exp\left[- \left(\sum_{j=1}^m \lambda_j \right) t \right] , \tag{7.11}$$

whose proof follows from theorem 3.6.

The conditional probability that a component in $\mathcal{C}$ is the cause of a system failure given that $\mathrm{S} = t$ and $\mathrm{W} = w$ is given by

$$p_{\mathcal{C}\,|\,\mathrm{S,W}}(\mathsf{C} \,|\, t, w, \boldsymbol{\lambda}^\star) = \frac{\sum_{j \in \mathsf{C}} \lambda_j^\star}{\binom{m-1}{w-1} \sum_{p=1}^m \lambda_p^\star} \mathbb{1}_{\mathrm{t}>0} \, \mathbb{1}_{|\mathsf{C}|=\mathrm{w}} \,. \tag{7.12}$$

whose proof follows from corollary 3.7.1.

Note that the marginal distribution of $\mathcal{C} \,|\, \mathrm{W} = w$ is given by

$$p_{\mathcal{C}\,|\,\mathrm{W}}(\mathsf{C} \,|\, w, \boldsymbol{\lambda}^\star) = \frac{\sum_{j \in \mathsf{C}} \lambda_j^\star}{\binom{m-1}{w-1} \sum_{p=1}^m \lambda_p^\star} \mathbb{1}_{|\mathsf{C}|=\mathrm{w}} \,, \tag{7.13}$$

Proof. The marginal is given by

$$p_{\mathcal{C}\,|\,\mathrm{W}}(\mathsf{C}\,|\,w, \boldsymbol{\lambda}^\star) = \int_{-\infty}^{\infty} f_{\mathcal{C},\mathrm{S}\,|\,\mathrm{W}}\left(\mathsf{C}, t\,|\,w, \boldsymbol{\lambda}^\star\right) \mathrm{d}t \tag{a}$$

$$= \int_0^{\infty} \frac{\mathbb{1}_{t>0}\,\mathbb{1}_{|\mathsf{C}|=\mathrm{w}}}{\binom{m-1}{w-1}} \left(\textstyle\sum_{j\in\mathsf{C}} \lambda_j\right) \exp\left[-\left(\textstyle\sum_{j=1}^m \lambda_j\right) t\right] \mathrm{d}t \tag{b}$$

$$= \frac{\mathbb{1}_{|\mathsf{C}|=\mathrm{w}}}{\binom{m-1}{w-1}} \sum_{j\in\mathsf{C}} \lambda_j \int_0^{\infty} \exp\left[-\left(\textstyle\sum_{j=1}^m \lambda_j\right) t\right] \mathrm{d}t\,. \tag{c}$$

Multiplying the integrand by

$$\frac{\sum_{j=1}^m \lambda_j}{\sum_{j=1}^m \lambda_j} \tag{d}$$

results in

$$p_{\mathcal{C}\,|\,\mathrm{W}}(\mathsf{C}\,|\,w, \boldsymbol{\lambda}^\star) = \frac{\mathbb{1}_{|\mathsf{C}|=\mathrm{w}}}{\binom{m-1}{w-1}} \frac{\sum_{j\in\mathsf{C}} \lambda_j}{\sum_{j=1}^m \lambda_j} \int_0^{\infty} \left(\textstyle\sum_{j=1}^m \lambda_j\right) \exp\left[-\left(\textstyle\sum_{j=1}^m \lambda_j\right) t\right] \mathrm{d}t \tag{e}$$

$$= \frac{\mathbb{1}_{|\mathsf{C}|=\mathrm{w}}}{\binom{m-1}{w-1}} \frac{\sum_{j\in\mathsf{C}} \lambda_j}{\sum_{j=1}^m \lambda_j} \int_0^{\infty} f_{\mathrm{S}}\left(t\,|\,\boldsymbol{\lambda}^\star\right) \mathrm{d}t \tag{f}$$

$$= \frac{\mathbb{1}_{|\mathsf{C}|=\mathrm{w}}}{\binom{m-1}{w-1}} \frac{\sum_{j\in\mathsf{C}} \lambda_j}{\sum_{j=1}^m \lambda_j}\,. \tag{g}$$

□

The marginal distribution of $\mathcal{C}\,|\,\mathrm{W} = w$ given by **??** is the same as the joint distribution of $\mathcal{C}, \mathrm{S}\,|\,\mathrm{W} = w$ given by eq. (7.13). Thus, $\mathcal{C}$ and S are conditionally independent given $\mathrm{W} = w$.

7.2 The maximum entropy distribution

The larger the variance of the estimator, the larger a sample necessary to obtain an estimate with a given mean squared error.

Entropy is a fundamental measure of uncertainty. If a diagnostician could ask "yes" or "no" questions about the component cause of failure, an *expected lower-bound* on the number of questions required to determine which component caused the failure is given by

$$\mathrm{H}(\mathrm{K}) = -\sum_{k\in\mathsf{K}} p_{\mathrm{K}}(k) \log_2 p_{\mathrm{K}}(k)\,. \tag{7.14}$$

This is known as the *entropy* of the discrete random variable K, which quantifies the amount of uncertainty about which value K will realize.

The greater the entropy, the greater the number of questions the diagnostician needs on average to determine the component cause of failure. Each component may have a different distribution of lifetimes, and thus knowing when a system failure occurs may *reduce* the uncertainty about the component cause of failure,

$$\mathrm{H}(\mathrm{K}\,|\,\mathrm{S}) \leq \mathrm{H}(\mathrm{K})\,. \tag{7.15}$$

If K and S are *statistically independent*, then

$$\mathrm{H}(\mathrm{K}\,|\,\mathrm{S}) = \mathrm{H}(\mathrm{K})\,, \tag{7.16}$$

that is, observing the system failure time does not reduce the uncertainty. The joint entropy of K and S is given by
$$\mathrm{H}(\mathrm{K},\mathrm{S}) = -\sum_{k \in \mathsf{K}} f_{\mathrm{K},\mathrm{S}}(k,t) \log_2 f_{\mathrm{K},\mathrm{S}}(k,t)\,, \tag{7.17}$$
which is equivalent to
$$\mathrm{H}(\mathrm{K} \mid \mathrm{S}) + \mathrm{H}(\mathrm{S})\,. \tag{7.18}$$

Given that $\mathrm{H}(\mathrm{K} \mid \mathrm{S}) \leq \mathrm{H}(\mathrm{K})$, a maximum entropy distribution has the form

$$\mathrm{H}(\mathrm{K},\mathrm{S}) = \mathrm{H}(\mathrm{K}) + \mathrm{H}(\mathrm{S})\,, \tag{7.19}$$

where each of the components, and thus the system lifetime S, has an interval of support $(0,\infty)$. We may thus maximize each part above, H(S) and H(K), independently, to derive the maximum entropy.

If we constrain the system lifetime to have an expectation μ (which may be easily estimable from a sample), then the constrained maximum entropy distribution of S is given by

$$\mathrm{S} \sim \mathrm{EXP}(\lambda) \tag{7.20}$$

where $\lambda = 1/\mu$.

Independently, the maximum entropy distribution of the discrete random variable K is given by the discrete uniform distribution

$$\mathrm{K} \sim \mathrm{UNIF}(1, m)\,, \tag{7.21}$$

which is fortunate since the only series system that has an exponentially distributed lifetime is a series system where each of the components has exponentially distributed lifetimes. Thus, each component must have a lifetime
$$\mathrm{T}_j \sim \mathrm{EXP}(\lambda/m) \tag{7.22}$$
for $j = 1, \ldots, m$, which is independent of S.

7.3 Maximum likelihood estimator

According to definition 5.3, the maximum likelihood estimator of $\boldsymbol{\lambda}^\star$ is the value that maximizes the log-likelihood on a given masked system failure time sample, denoted $\mathbf{M_n}$. The likelihood function is given by
$$\mathcal{L}(\boldsymbol{\lambda} \mid \mathbf{M_n}) \propto \exp\left[-\left(\sum_{j=1}^{m} \lambda_j\right)\left(\sum_{i=1}^{n} t_i\right)\right]\left[\prod_{i=1}^{n}\left(\sum_{j \in \mathsf{C}_i} \lambda_j\right)\right], \tag{7.23}$$

whose proof follows from definition 4.2. Therefore, the log-likelihood function ℓ is given by

$$\ell(\boldsymbol{\lambda} \mid \mathbf{M_n}) = \text{const} + \sum_{i=1}^{n} \ln\left[\sum_{j \in \mathsf{C}_i} \lambda_j\right] - \left[\sum_{i=1}^{n} t_i\right]\left[\sum_{j=1}^{m} \lambda_j\right]. \tag{7.24}$$

Thus, maximum likelihood estimator of $\boldsymbol{\lambda}^\star$ is given by

$$\hat{\boldsymbol{\lambda}}_{\mathbf{n}} = \operatorname*{arg\,max}_{\boldsymbol{\lambda} > \mathbf{0}} \ell(\boldsymbol{\lambda} \mid \mathbf{M_n})\,. \tag{7.25}$$

The solution to eq. (7.25) must be a stationary point, i.e., a point at which the score function is zero. The j^{th} component of the score function is given by

$$\frac{\partial \ell}{\partial \lambda_j} = \sum_{i=1}^{n} \left(\sum_{p \in \mathsf{C}_i} \lambda_p \right)^{-1} \mathbb{1}_{\mathrm{j} \in \mathsf{C}_\mathrm{i}} - \sum_{i=1}^{n} t_i \,. \tag{7.26}$$

According to postulate 5.4, the sampling distribution of $\hat{\boldsymbol{\lambda}}_{\mathbf{n}}$ converges in distribution to a multivariate normal with a mean given by $\boldsymbol{\lambda}^\star$ and a variance-covariance given by

$$\frac{1}{n} \boldsymbol{\mathcal{I}}^{-1}(\boldsymbol{\lambda}^\star) \,. \tag{7.27}$$

Since $\boldsymbol{\lambda}^\star$ is not known it may be approximated by

$$\boldsymbol{\mathcal{I}}^{-1}\left(\hat{\boldsymbol{\lambda}}_{\mathbf{n}}\right) \,. \tag{7.28}$$

The (j,k)-th element of the Fisher information matrix is given by

$$\left[\boldsymbol{\mathcal{I}}(\boldsymbol{\lambda}^\star)\right]_{jk} = \frac{\sum_{\mathsf{C}} \left[\left(\sum_{p \in \mathsf{C}} \lambda_p^\star \right)^{-1} \mathbb{1}_{\mathrm{j} \in \mathsf{C}, \mathrm{k} \in \mathsf{C}} \right]}{\binom{m-1}{w-1} \sum_{p=1}^{m} \lambda_p^\star} \,. \tag{7.29}$$

where the summation indexed by C is over $\mathcal{P}(\{1,\ldots,m\})$.

Proof. By eq. (4.11), the (j,k)-th element of the information matrix is given by

$$\left[\boldsymbol{\mathcal{I}}(\boldsymbol{\lambda}^\star)\right]_{ij} = -\sum_{\mathsf{C}} \int_0^\infty \frac{\partial^2}{\partial \lambda_i \partial \lambda_j} \ln f_{\mathcal{C},\mathrm{S}\,|\,\mathrm{W}}\left(\mathsf{C}, t \,|\, w, \operatorname{mat}(\boldsymbol{\lambda})\right)\bigg|_{\boldsymbol{\lambda}=\boldsymbol{\lambda}^\star} \cdot f_{\mathcal{C},\mathrm{S}\,|\,\mathrm{W}}\left(\mathsf{C}, t \,|\, w, \operatorname{mat}(\boldsymbol{\lambda}^\star)\right) \mathrm{d}t \,, \tag{a}$$

In the above equation, the second-order partial derivative of the log of the joint density is equivalent to the partial derivative with respect to the k^{th} parameter of the j^{th} component of the score, given by eq. (7.26), conditioned on a masked system failure time sample of size 1, denoted by $\mathbf{M}_{\mathbf{n}1}$. This mixed second-order partial derivative is given by

$$\frac{\partial}{\partial \lambda_k} \frac{\partial \ell}{\partial \lambda_j} = \frac{\partial}{\partial \lambda_k} \left(\left[\sum_{p \in \mathsf{C}} \lambda_p \right]^{-1} \mathbb{1}_{\mathrm{j} \in \mathsf{C}} - t \right) \tag{b}$$

$$= -\left(\sum_{p \in \mathsf{C}} \lambda_p \right)^{-2} \mathbb{1}_{\mathrm{j} \in \mathsf{C}, \mathrm{k} \in \mathsf{C}} \,. \tag{c}$$

Plugging in the joint density and substituting eq. (c) into eq. (a) results in

$$\left[\boldsymbol{\mathcal{I}}(\boldsymbol{\lambda}^\star)\right]_{ij} = \sum_{\mathsf{C}} \int_0^\infty \left(\sum_{p \in \mathsf{C}} \lambda_p \right)^{-2} \mathbb{1}_{\mathrm{p} \in \mathsf{C}, \mathrm{k} \in \mathsf{C}} \bigg|_{\boldsymbol{\lambda}=\boldsymbol{\lambda}^\star} \frac{1}{\binom{m-1}{w-1}} \left(\sum_{p \in \mathsf{C}} \lambda_p^\star \right) \exp\left(-\left[\sum_{p=1}^{m} \lambda_p^\star \right] t \right) \mathrm{d}t \,. \tag{d}$$

Simplifying and rearranging results in

$$\left[\boldsymbol{\mathcal{I}}(\boldsymbol{\lambda}^\star)\right]_{ij} = \frac{1}{\binom{m-1}{w-1}} \sum_{\mathsf{C}} \left(\sum_{p \in \mathsf{C}} \lambda_p^\star \right)^{-1} \mathbb{1}_{\mathrm{p} \in \mathsf{C}, \mathrm{k} \in \mathsf{C}} \int_0^\infty \exp\left(-\left[\sum_{p=1}^{m} \lambda_p^\star \right] t \right) \mathrm{d}t \,. \tag{e}$$

Multiplying the integral by $\frac{\sum_{p=1}^m \lambda_p^\star}{\sum_{p=1}^m \lambda_p^\star}$ results in an integral of the form

$$\int_0^\infty \frac{\sum_{p=1}^m \lambda_p^\star}{\sum_{p=1}^m \lambda_p^\star} \exp\left(-\left[\sum_{p=1}^m \lambda_p^\star\right] t\right) \mathrm{d}t\,. \tag{f}$$

Pulling the $\left(\sum_{p=1}^m \lambda_p^\star\right)^{-1}$ constant out of the integrand results in

$$\frac{1}{\sum_{p=1}^m \lambda_p^\star} \int_0^\infty \left(\sum_{p=1}^m \lambda_p^\star\right) \exp\left(-\left[\sum_{p=1}^m \lambda_p^\star\right] t\right) \mathrm{d}t = \frac{1}{\sum_{p=1}^m \lambda_p^\star} \int_0^\infty f_{\mathsf{S}}(t \mid \boldsymbol{\lambda}^\star)\,\mathrm{d}t \tag{g}$$

$$= \frac{1}{\sum_{p=1}^m \lambda_p^\star}\,. \tag{h}$$

Substituting eq. (h) into eq. (e) results in

$$\left[\boldsymbol{\mathcal{I}}(\boldsymbol{\lambda}^\star)\right]_{ij} = \frac{1}{\binom{m-1}{w-1}} \sum_{\mathsf{C}} \left\{ \left(\sum_{p \in \mathsf{C}} \lambda_p^\star\right)^{-1} \frac{1}{\sum_{p=1}^m \lambda_p^\star} \mathbb{1}_{\mathrm{p}\in\mathsf{C},\mathrm{k}\in\mathsf{C}} \right\}. \tag{i}$$

□

7.3.1 Sufficient statistics

Consider a sample of n masked system failure times denoted by $\mathbf{m_n}$. The mean system lifetime of the sample is given by

$$\bar{t} = \frac{1}{n}\sum_{i=1}^n \mathrm{t_i}\,. \tag{7.30}$$

and another statistic, denoted by $\hat{\boldsymbol{\omega}}$, is a dictionary of candidate set-frequency count pairs where a candidate set $\mathbf{k} \in \mathcal{P}(\mathsf{K})$ maps to its sample frequency and is given by

$$\hat{\boldsymbol{\omega}}_{\mathbf{k}} = \sum_{i=1}^n \mathbb{1}_{\mathbf{k}}(\mathsf{C}_i)\,. \tag{7.31}$$

Given a sample $\mathbf{m_n}$, the likelihood $\mathcal{L}(\boldsymbol{\lambda} \mid \mathbf{m_n})$ described by eq. (7.23) is the same as the likelihood given by

$$\mathcal{L}(\boldsymbol{\lambda} \mid \bar{t}, \hat{\boldsymbol{\omega}}) = \exp\left(-n\bar{t}\sum_{j=1}^m \lambda_j\right) \prod_{\mathsf{C}\in\mathcal{W}_w} \left(\sum_{j\in\mathsf{C}} \lambda_j\right)^{\hat{\boldsymbol{\omega}}_{\mathsf{C}}}, \tag{7.32}$$

where $\mathcal{W}_w$ is the set of all candidate sets of cardinality w.

Proof. The likelihood function $\mathcal{L}$ with respect to a sample of n masked system failure times is given by

$$\mathcal{L}(\boldsymbol{\lambda} \mid \mathbf{m_n}) \propto \exp\left[-\left(\sum_{j=1}^m \lambda_j\right)\left(\sum_{i=1}^n t_i\right)\right]\left[\prod_{i=1}^n \left(\sum_{j\in\mathsf{C}_i} \lambda_j\right)\right]. \tag{7.23 revisited}$$

Substituting $\sum_{i=1}^{n} \mathrm{t_i}$ in the above equation with the equivalent expression $n\bar{t}$ results in

$$\mathcal{L}(\boldsymbol{\lambda} \,|\, \mathbf{M_n}) \propto \exp\left[-n\bar{t}\sum_{j=1}^{m}\lambda_j\right]\underbrace{\left[\prod_{i=1}^{n}\left(\sum_{j\in \mathsf{C}_i}\lambda_j\right)\right]}_{A}. \tag{a}$$

In the above equation, each unique set $(\lambda_{j_1} + \cdots + \lambda_{j_w})$ in the product (labeled A) has multiplicity $\hat{\boldsymbol{\omega}}_{\{\lambda_{j_1}\cdots\lambda_{j_w}\}}$, thus we may substitute this product with the equivalent product

$$\prod_{\mathsf{C}\in\mathcal{W}_w}\left(\sum_{j\in\mathsf{C}}\lambda_j\right)^{\hat{\omega}_{\mathsf{C}}}, \tag{b}$$

where C is over all unique candidate sets of cardinality w. The result of this substitution is given by

$$\mathcal{L}(\boldsymbol{\lambda} \,|\, \bar{t}, \hat{\boldsymbol{\omega}}) = \exp\left(-n\bar{t}\sum_{j=1}^{m}\lambda_j\right)\prod_{\mathsf{C}\in\mathcal{W}_w}\left(\sum_{j\in\mathsf{C}}\lambda_j\right)^{\hat{\omega}_{\mathsf{C}}}. \tag{c}$$

□

The statistics $\bar{t}$ and $\hat{\boldsymbol{\omega}}$ are jointly sufficient for $\boldsymbol{\lambda}^{\star}$

Proof. By the Fisher–Neyman factorization theorem, since $\mathcal{L}$ given by eq. (7.23) can be factored as $\mathcal{L}(\bar{t}, \hat{\boldsymbol{\omega}}) = h(\bar{t}, \hat{\boldsymbol{\omega}})g(\bar{t}, \hat{\boldsymbol{\omega}})$, $\hat{\boldsymbol{\omega}}$ and $\bar{t}$ are joint sufficient statistics for $\boldsymbol{\lambda}^{\star}$. □

It may be helpful to consider an example given by the following

Example 1 *Let a masked failure time sample of size n for a 3-out-of-3 system with exponentially distributed lifetime components be given by*

$$\mathbf{m_n}4 = \Big((t_1 = 1, \mathsf{C}_1 = \{1,2\}), (t_2 = 1, \mathsf{C}_2 = \{1,3\}), (t_3 = 2, \mathsf{C}_3 = \{1,2\}), (t_4 = 2, \mathsf{C}_4 = \{1,2\})\Big).$$

Then, joint sufficient statics of $\boldsymbol{\lambda}^{\star}$ are given by

$$\bar{t} = \frac{1+1+2+2}{4} = \frac{3}{2},$$
$$\hat{\boldsymbol{\omega}} = \{\hat{\boldsymbol{\omega}}_{\{1,2\}} \mapsto 3, \hat{\boldsymbol{\omega}}_{\{1,3\}} \mapsto 1\}.$$

No additional information is needed to compute the likelihood of $\boldsymbol{\lambda}$ with respect to the given $\mathbf{M_{n4}}$ sample.

The log-likelihood ℓ is given by

$$\ell(\boldsymbol{\theta} \,|\, \bar{t}, \hat{\boldsymbol{\omega}}, w) = -n\bar{t}\left(\sum_{j=1}^{m}\lambda_j\right) + \sum_{\mathsf{C}}\hat{\boldsymbol{\omega}}_{\mathsf{C}}\ln\left(\sum_{j\in\mathsf{C}}\lambda_j\right). \tag{7.33}$$

To find the MLE $\hat{\boldsymbol{\lambda}}_n$, we find the stationary points for $\lambda_1, \ldots, \lambda_m$ given by

$$0 = -n\bar{t} + \sum_{\mathsf{C}}\hat{\boldsymbol{\omega}}_{\mathsf{C}}\left(\sum_{j\in\mathsf{C}}\lambda_j\right)^{-1}\mathbb{1}_{\mathrm{k}\in\mathsf{C}}\,; k = 1, \ldots, m. \tag{7.34}$$

The (j,k)-th element of the observed information matrix $\boldsymbol{\mathcal{J}}_n$ is given by

$$\left[\boldsymbol{\mathcal{J}}_n(\boldsymbol{\lambda}^\star \hat{\boldsymbol{\omega}})\right]_{jk} = -\sum_{\mathsf{C}} \hat{\boldsymbol{\omega}}_{\mathsf{C}} \left(\sum_{i \in \mathsf{C}} \lambda_i^\star\right)^{-2} \mathbb{1}_{\mathsf{j}\in\mathsf{C}}\, \mathbb{1}_{\mathsf{k}\in\mathsf{C}}\,, \tag{7.35}$$

which only depends on the statistic $\hat{\boldsymbol{\omega}}$.

7.4 Applications

Knowing the distribution of component lifetimes has many important applications. The canonical use-case is, if a failure occurs in a series system at time t, for any component we know the probability that it is the component cause of failure. However, there are other applications as well. For instance, a system designer may desire a system where the component cause of failure is as uncertain as possible to prevent an *adversary* from learning about critical vulnerabilities.

A $(1-\alpha)\cdot 100\%$-confidence interval for $\lambda_j^\star$ is given by

$$\hat{\lambda}_j \pm z_{1-\alpha/2}\sqrt{\frac{1}{n}\left[\boldsymbol{\mathcal{I}}^{-1}(\hat{\boldsymbol{\lambda}}_{\mathbf{n}})\right]_{jj}}\,. \tag{7.36}$$

The expected lifetime of the system is given by

$$\mathbb{E}_{\boldsymbol{\lambda}^\star}[\mathrm{S}] = \frac{1}{\sum_{p=1}^m \lambda_j^\star}\,. \tag{7.37}$$

The maximum likelihood estimator of $\mathbb{E}_{\boldsymbol{\lambda}^\star}[\mathrm{S}]$ is given by $\mathbb{E}_{\hat{\boldsymbol{\lambda}}_{\mathbf{n}}}[\mathrm{S}]$, which has an asymptotic sampling distribution given by

$$\mathrm{S}_{\mathbb{E}_{\hat{\boldsymbol{\lambda}}_{\mathbf{n}}}[\mathrm{S}]} \sim \mathcal{N}\left(\mathbb{E}_{\boldsymbol{\lambda}^\star}[\mathrm{S}], \mathrm{Var}\left[\mathbb{E}_{\hat{\boldsymbol{\lambda}}_{\mathbf{n}}}[\mathrm{S}]\right]\right)\,, \tag{7.38}$$

where

$$\mathrm{Var}\left[\mathbb{E}_{\hat{\boldsymbol{\lambda}}_{\mathbf{n}}}[\mathrm{S}]\right] = a\,. \tag{7.39}$$

If a system failure occurs, the expected number of inspections is given by

$$\mathbb{E}_{\boldsymbol{\lambda}^\star}[\mathrm{N}] = \frac{\sum_{n=1}^m n\cdot\lambda_{\pi(n)}^\star}{\sum_{p=1}^m \lambda_p^\star}\,, \tag{7.40}$$

where $\pi(j) \leq \pi(k) \implies \lambda_j^\star \geq \lambda_k^\star$. The maximum likelihood estimator of $\mathbb{E}_{\boldsymbol{\lambda}^\star}[\mathrm{N}]$ is given by $\mathbb{E}_{\hat{\boldsymbol{\lambda}}_{\mathbf{n}}}[\mathrm{N}]$, which has an asymptotic sampling distribution given by

$$\mathrm{S}_{\mathbb{E}_{\hat{\boldsymbol{\lambda}}_{\mathbf{n}}}[\mathrm{N}]} \sim \mathcal{N}\left(\mathbb{E}_{\boldsymbol{\lambda}^\star}[\mathrm{N}], \mathrm{Var}\left[\mathbb{E}_{\hat{\boldsymbol{\lambda}}_{\mathbf{n}}}[\mathrm{N}]\right]\right)\,, \tag{7.41}$$

where an estimator of the variance-covariance $\mathrm{Var}\left[\mathbb{E}_{\hat{\boldsymbol{\lambda}}_{\mathbf{n}}}[\mathrm{N}]\right]$ is described in section 6.3.

7.5 Case study: 3-out-of-3 system

The 3-out-of-3 system's lifetime distribution functions are given by

$$R_{\mathrm{S}}(t \mid \boldsymbol{\lambda}^\star) = \exp\left[-(\lambda_1^\star + \lambda_2^\star + \lambda_3^\star)t\right]\,, \tag{7.42}$$

$$f_{\mathrm{S}}(t \mid \boldsymbol{\lambda}^\star) = (\lambda_1^\star + \lambda_2^\star + \lambda_3^\star)\exp\left[-(\lambda_1^\star + \lambda_2^\star + \lambda_3^\star)t\right]\,, \tag{7.43}$$

$$h_{\mathrm{S}}(t \mid \boldsymbol{\lambda}^\star) = \lambda_1^\star + \lambda_2^\star + \lambda_3^\star\,, \tag{7.44}$$

where $t > 0$, $\lambda_1^\star > 0$, $\lambda_2^\star > 0$, and $\lambda_3^\star > 0$.

7.5.1 Sampling distributions

Two candidates

If we condition on samples in which each observation consists of two candidates, $\mathrm{W} = |\mathcal{C}| = 2$, the likelihood function is given by

$$\mathcal{L}(\boldsymbol{\lambda} \,|\, \bar{t}, \hat{\boldsymbol{\omega}}) = (\lambda_1 + \lambda_2)^{\hat{\omega}_{\{1,2\}}} (\lambda_1 + \lambda_3)^{\hat{\omega}_{\{1,3\}}} (\lambda_2 + \lambda_3)^{\hat{\omega}_{\{2,3\}}} \exp\left(-n\bar{t}\,(\lambda_1 + \lambda_2 + \lambda_3)\right) \tag{7.45}$$

and the log-likelihood function is given by

$$\begin{aligned}\ell(\boldsymbol{\lambda} \,|\, \bar{t}, \hat{\boldsymbol{\omega}}) = \hat{\boldsymbol{\omega}}_{\{1,2\}} \ln(\lambda_1 + \lambda_2) + \hat{\boldsymbol{\omega}}_{\{1,3\}} \ln(\lambda_1 + \lambda_3) + \\ \hat{\boldsymbol{\omega}}_{\{2,3\}} \ln(\lambda_2 + \lambda_3) - n\bar{t}(\lambda_1 + \lambda_2 + \lambda_3)\,.\end{aligned} \tag{7.46}$$

The maximum likelihood estimate $\hat{\boldsymbol{\lambda}}_{\mathbf{n}}$ is given by the solving the system of equations in which the score is zero, $\mathrm{s}(\boldsymbol{\lambda}) = \mathbf{0}$, where the score is given by

$$\mathrm{s}\left(\boldsymbol{\lambda}\right) = \begin{pmatrix} \frac{\hat{\omega}_{\{1,2\}}}{\lambda_1+\lambda_2} + \frac{\hat{\omega}_{\{1,3\}}}{\lambda_1+\lambda_3} \\ \frac{\hat{\omega}_{\{1,2\}}}{\lambda_1+\lambda_2} + \frac{\hat{\omega}_{\{2,3\}}}{\lambda_2+\lambda_3} \\ \frac{\hat{\omega}_{\{1,3\}}}{\lambda_1+\lambda_3} + \frac{\hat{\omega}_{\{2,3\}}}{\lambda_1+\lambda_3} \end{pmatrix} - n\bar{t} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\,. \tag{7.47}$$

This has a closed form solution given by

$$\hat{\boldsymbol{\lambda}}_{\mathbf{n}} = \frac{1}{n\bar{t}} \begin{pmatrix} \hat{\boldsymbol{\omega}}_{\{1,2\}} + \hat{\boldsymbol{\omega}}_{\{1,3\}} - \hat{\boldsymbol{\omega}}_{\{2,3\}} \\ \hat{\boldsymbol{\omega}}_{\{1,2\}} - \hat{\boldsymbol{\omega}}_{\{1,3\}} + \hat{\boldsymbol{\omega}}_{\{2,3\}} \\ -\hat{\boldsymbol{\omega}}_{\{1,2\}} + \hat{\boldsymbol{\omega}}_{\{1,3\}} + \hat{\boldsymbol{\omega}}_{\{2,3\}} \end{pmatrix}\,. \tag{7.48}$$

The information matrix is given by

$$\boldsymbol{\mathcal{I}}\left(\boldsymbol{\lambda}^\star\right) = \frac{1}{2(\lambda_1^\star + \lambda_2^\star + \lambda_3^\star)} \begin{bmatrix} \frac{1}{\lambda_1^\star+\lambda_2^\star} + \frac{1}{\lambda_1^\star+\lambda_3^\star} & \frac{1}{\lambda_1^\star+\lambda_2^\star} & \frac{1}{\lambda_1^\star+\lambda_3^\star} \\ \frac{1}{\lambda_1^\star+\lambda_2^\star} & \frac{1}{\lambda_1^\star+\lambda_2^\star} + \frac{1}{\lambda_2^\star+\lambda_3^\star} & \frac{1}{\lambda_2^\star+\lambda_3^\star} \\ \frac{1}{\lambda_1^\star+\lambda_3^\star} & \frac{1}{\lambda_2^\star+\lambda_3^\star} & \frac{1}{\lambda_1^\star+\lambda_3^\star} + \frac{1}{\lambda_2^\star+\lambda_3^\star} \end{bmatrix} \tag{7.49}$$

To derive the variance-covariance of the sampling distribution of $\hat{\boldsymbol{\lambda}}_{\mathbf{n}}$ for a sample of n masked system failure times, denoted by $\boldsymbol{\Lambda}_{\mathbf{n}}$, we take the inverse of $n \cdot \boldsymbol{\mathcal{I}}\left(\boldsymbol{\lambda}^\star\right)$ resulting in

$$\mathrm{Var}_{\boldsymbol{\lambda}^\star}[\boldsymbol{\Lambda}_{\mathbf{n}}] = \frac{\lambda_1^\star + \lambda_2^\star + \lambda_3^\star}{n} \begin{bmatrix} \lambda_1^\star + \lambda_2^\star + \lambda_3^\star & -\lambda_3^\star & -\lambda_2^\star \\ -\lambda_3^\star & \lambda_1^\star + \lambda_2^\star + \lambda_3^\star & -\lambda_1^\star \\ -\lambda_2^\star & -\lambda_1^\star & \lambda_1^\star + \lambda_2^\star + \lambda_3^\star \end{bmatrix}\,. \tag{7.50}$$

By the asymptotic unbiasedness of $\hat{\boldsymbol{\lambda}}_{\mathbf{n}}$, the asymptotic mean squared error is given by the trace of the variance-covariance matrix,

$$\mathrm{MSE}\left(\boldsymbol{\Lambda}_{\mathbf{n}}\right) = \frac{3(\lambda_1^\star + \lambda_2^\star + \lambda_3^\star)^2}{n}\,. \tag{7.51}$$

According to theorem 5.1, for a sufficiently large sample size n,

$$\boldsymbol{\Lambda}_{\mathbf{n}} \sim \mathrm{MVN}\left(\boldsymbol{\lambda}^\star, \frac{1}{n}\boldsymbol{\mathcal{I}}^{-1}(\boldsymbol{\lambda}^\star)\right)\,. \tag{7.52}$$

Consequently, each time we observe a particular $\boldsymbol{\Lambda}_{\mathbf{n}} = \hat{\boldsymbol{\lambda}}_{\mathbf{n}}$, we draw a random vector from the multivariate normal distribution given by eq. (7.52). Thus, the independent asymptotic $(1-\alpha)\cdot 100\%$-confidence interval for $\lambda_j^\star$ is given by

$$\hat{\lambda}_j \pm \frac{z_{1-\alpha/2}}{\sqrt{n}} \left(\hat{\lambda}_1 + \hat{\lambda}_2 + \hat{\lambda}_3\right)\,. \tag{7.53}$$

One candidate

If we condition on samples in which each observation consists of one candidate, $\mathrm{W} = |\mathcal{C}| = 1$, the likelihood function is given by

$$\mathcal{L}(\boldsymbol{\lambda} \,|\, \bar{t}, \hat{\boldsymbol{\omega}}) = \lambda_1^{\hat{\omega}\{1\}} \lambda_2^{\hat{\omega}\{2\}} \lambda_3^{\hat{\omega}\{3\}} \exp\left(-n\bar{t}\,(\lambda_1 + \lambda_2 + \lambda_3)\right) \tag{7.54}$$

and the log-likelihood function is given by

$$\ell(\boldsymbol{\lambda} \,|\, \bar{t}, \hat{\boldsymbol{\omega}}) = \hat{\boldsymbol{\omega}}_{\{1\}} \ln \lambda_1 + \hat{\boldsymbol{\omega}}_{\{2\}} \ln \lambda_2 + \hat{\boldsymbol{\omega}}_{\{3\}} \ln \lambda_3 - n\bar{t}\,(\lambda_1 + \lambda_2 + \lambda_3)\ . \tag{7.55}$$

The maximum likelihood estimator is given by

$$\hat{\boldsymbol{\lambda}}_{\mathbf{n}} = \frac{1}{n\bar{t}} \begin{pmatrix} \hat{\boldsymbol{\omega}}_{\{1\}} \\ \hat{\boldsymbol{\omega}}_{\{2\}} \\ \hat{\boldsymbol{\omega}}_{\{3\}} \end{pmatrix}, \tag{7.56}$$

and the information matrix is given by

$$\boldsymbol{\mathcal{I}}\left(\boldsymbol{\lambda}^\star\right) = \frac{1}{\lambda_1^\star + \lambda_2^\star + \lambda_3^\star} \begin{bmatrix} \frac{1}{\lambda_1^\star} & 0 & 0 \\ 0 & \frac{1}{\lambda_2^\star} & 0 \\ 0 & 0 & \frac{1}{\lambda_3^\star} \end{bmatrix}, \tag{7.57}$$

and thus the variance-covariance matrix is given by

$$\Sigma_n\left(\boldsymbol{\lambda}^\star\right) = \frac{\lambda_1^\star + \lambda_2^\star + \lambda_3^\star}{n} \begin{bmatrix} \lambda_1^\star & 0 & 0 \\ 0 & \lambda_2^\star & 0 \\ 0 & 0 & \lambda_3^\star \end{bmatrix}. \tag{7.58}$$

The asymptotic mean squared error is given by the trace of the variance-covariance matrix,

$$\mathrm{MSE}\left(\boldsymbol{\Lambda}_{\mathbf{n}}\right) = \frac{(\hat{\lambda}_1 + \hat{\lambda}_2 + \hat{\lambda}_3)^2}{n}, \tag{7.59}$$

which has three times less mean squared error than when conditioning on $|\mathcal{C}| = 2$. That is, an $\mathbf{M_n}$ sample in which each observation consists of one candidate has more information about $\boldsymbol{\lambda}^\star$ than a sample in which each observation consists of two candidates.

The independent asymptotic $(1-\alpha)\cdot 100\%$-confidence interval for $\lambda_j^\star$ is given by

$$\hat{\lambda}_j \pm z_{1-\alpha/2} \sqrt{\frac{\hat{\lambda}_j\left(\hat{\lambda}_1 + \hat{\lambda}_2 + \hat{\lambda}_3\right)}{n}}\,. \tag{7.60}$$

By comparison, the confidence interval when conditioning on $|\mathcal{C}| = 2$ is

$$\sqrt{\frac{\hat{\lambda}_1 + \hat{\lambda}_2 + \hat{\lambda}_3}{\hat{\lambda}_j}} \tag{7.61}$$

as wide.

Three candidates

The degenerate case is all components are candidates, $\mathrm{W} = |\mathcal{C}| = 3$. There is no maximum likelihood estimator $\hat{\boldsymbol{\lambda}}_{\mathbf{n}}$. Rather, solving the equation results in the underspecified system given by

$$\sum_{j=1}^{3} \hat{\lambda}_j - \frac{1}{\bar{t}} = 0\,, \tag{7.62}$$

where $\hat{\lambda}_j > 0$ for $j = 1, 2, 3$.

The solution set of this underspecified system is the equation of a plane bounded in the positive region as depicted in fig. 1. The plane has a normal vector given by

$$\mathbf{n} = \langle 1, 1, 1 \rangle^{\mathsf{T}}\,, \tag{7.63}$$

which is independent of the statistic $\bar{t}$. Across multiple samples, the statistic $\bar{t}$ varies but the normal vector of the plane does not. Thus, the sampling distribution of the solution set will be a density over planes bounded in the positive region with a normal vector given by eq. (7.63).

As n goes to infinity, the probability that $\boldsymbol{\lambda}^{\star}$ intersects the plane goes to 1. This information may not be very useful, especially if the system lifetime S has a small expected value (the area of the bounded plane is $1/\bar{t}^2$), but if this is the only information available, what is the best course of action? An unbiased estimator that asymptotically converges to $\boldsymbol{\lambda}^{\star}$ is not possible, but we can minimize some other loss function. If we assume $\boldsymbol{\lambda}^{\star}$ takes on any supported value with equal likelihood, then the estimate that minimizes the expected Euclidean distance from the projection of $\boldsymbol{\lambda}^{\star}$ onto the plane of the solution set is given by

$$\tilde{\boldsymbol{\lambda}}_{\mathbf{n}} = \frac{1}{3\bar{t}}\mathbf{n}\,. \tag{7.64}$$

7.5.2 Rate of convergence to the asymptotic sampling distribution

In this section, we evaluate how rapidly the sampling distribution of $\hat{\boldsymbol{\theta}}(n \,|\, w)$ converges to the asymptotic distribution as given by **??**, a multivariate normal with a mean $\boldsymbol{\theta}^{\star}$ and a variance-covariance matrix $\frac{1}{n}\boldsymbol{\mathcal{I}}^{-1}(\boldsymbol{\theta}^{\star})$.

By eq. (5.16), the asymptotic mean squared error of the maximum likelihood estimator $\hat{\boldsymbol{\theta}}(n \,|\, w)$ is given by the trace of the variance-covariance matrix, $\mathrm{MSE} = \frac{1}{n}\operatorname{tr}\left(\boldsymbol{\mathcal{I}}^{-1}\left(\boldsymbol{\theta}^{\star}\right)\right)$. An estimator of the mean squared error from a sample of r maximum likelihood estimates is given by

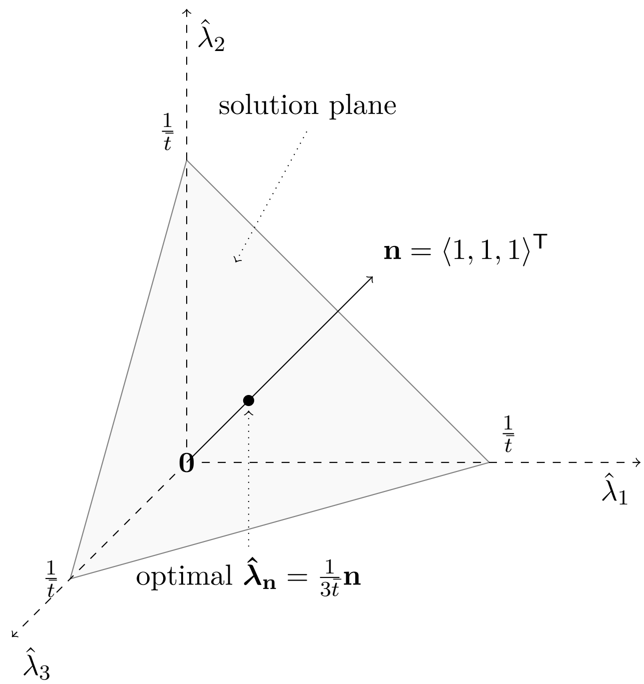
$$\hat{\mathrm{MSE}}\left(\hat{\boldsymbol{\theta}}^{(1)}(n \,|\, w), \ldots, \hat{\boldsymbol{\theta}}^{(r)}(n \,|\, w)\right) = \frac{1}{r}\sum_{i=1}^{r}\left(\hat{\boldsymbol{\theta}}^{(i)}(n \,|\, w) - \boldsymbol{\theta}^{\star}\right)^{\mathsf{T}}\left(\hat{\boldsymbol{\theta}}^{(i)}(n \,|\, w) - \boldsymbol{\theta}^{\star}\right). \tag{7.65}$$

By generating r masked system failure time of samples of size n, denoted by $\mathbf{M}(n, 1), \ldots, \mathbf{M}(r, 1)$, and finding the corresponding maximum likelihood estimates, denoted by $\hat{\boldsymbol{\theta}}^{(1)}(n \,|\, w), \ldots, \hat{\boldsymbol{\theta}}^{(r)}(n \,|\, w)$, we may estimate the mean squared error and compare the result against the asymptotic mean squared error. When we perform these steps for masked system failure time samples of size $n = 1, \ldots, N$, we may plot the rate of convergence.

The absolute difference between the mean squared error of the asymptotic covariance matrix and the "true" covariance matrix is given by

$$|\,\mathrm{MSE} - \hat{\mathrm{MSE}}|\,, \tag{7.66}$$

Figure 1: $\hat{\boldsymbol{\lambda}}_{\mathbf{n}}$ solutions given three candidates



*** Let's use a profile likelihood to look at how the likelihood function changes with respect to a parameter component. Say, for instance, the other cmoponents in the parameter are nuisance parameters, and we are really just interested in the j^{th} component. ***

Let $\boldsymbol{\lambda}^\star = \langle 2,3,4 \rangle^{\mathsf{T}}$ and let the sample of masked system failure times be of size $n = 1000$. Theoretically, the MLE $\hat{\boldsymbol{\lambda}}_{\mathbf{1000}}$ is approximately distrubted as

$$\hat{\boldsymbol{\lambda}}_{\mathbf{1000}} \sim \operatorname{MVN}\left(\langle 2,3,4 \rangle^{\mathsf{T}}, \boldsymbol{\Sigma}_{\mathbf{1000}}\right). \tag{7.67}$$

where

$$\boldsymbol{\Sigma}_{\mathbf{1000}} = \begin{pmatrix} 0.081 & -0.036 & -0.027 \\ -0.036 & 0.081 & -0.018 \\ -0.027 & -0.018 & 0.081 \end{pmatrix}. \tag{7.68}$$

We generated two different sets of data points, one from the theoretical distribution described in eq. (7.67) and the other from the maximum likelihood method applied to $\mathbf{M}_{\mathbf{n}1000}$ samples. Refer to fig. 2 for a visualization of the data points. Visually, they have a similar distribution of data points, indicating that the asymptotic distribution is a reasonable approximation of the sampling distribution of the MLE $\hat{\boldsymbol{\lambda}}_{1000}$.

A sample variance-covariance matrix of $r = 10000$ maximum likelihood estimates was computed to be

$$\widehat{\boldsymbol{\Sigma}_{\mathbf{1000}}} = \begin{pmatrix} 0.081 & -0.037 & -0.027 \\ -0.037 & 0.082 & -0.018 \\ -0.027 & -0.018 & 0.081 \end{pmatrix}, \tag{7.69}$$

which is approximately equivalent to the theoretical asymptotic distribution given by eq. (7.67).

To derive an accurate estimate of the sampling distribution of $\hat{\boldsymbol{\lambda}}_{\mathbf{n}}$ for small n, as opposed to the asymptotic sampling distribution for large n, we generate r masked system failure time samples

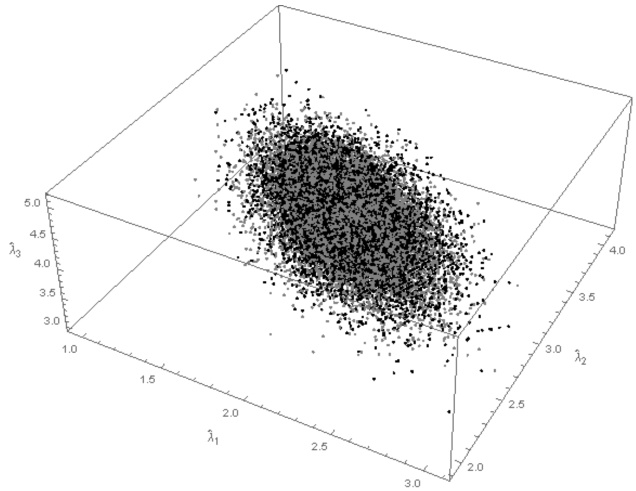
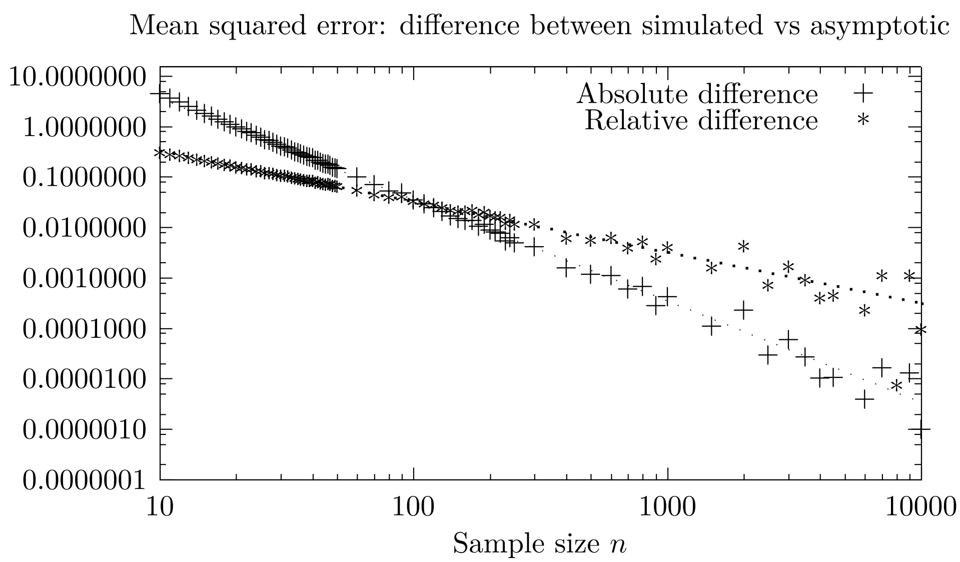


Figure 2: Theoretical MLE data points (gray) vs actual MLE data points (black)



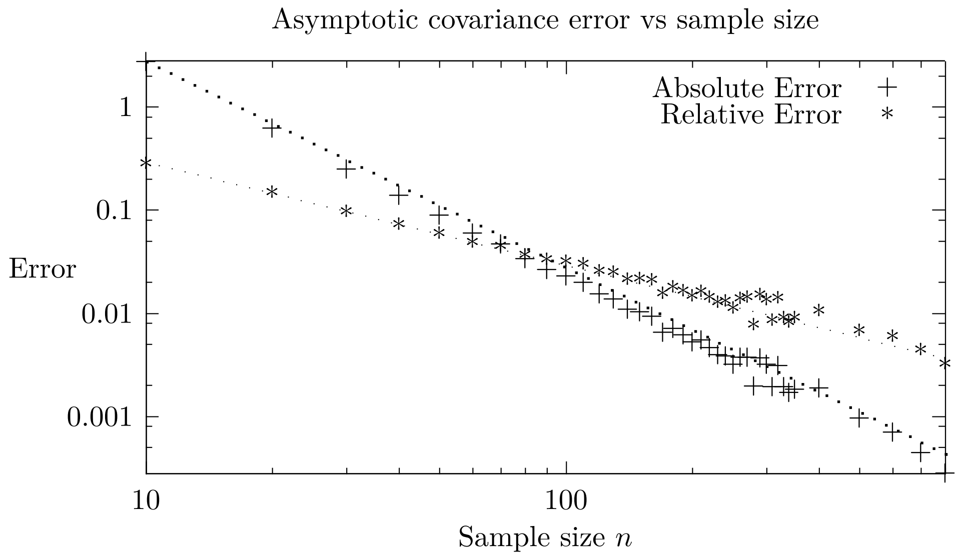
We define the error between two matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ to be given by

$$\|\boldsymbol{A}-\boldsymbol{B}\|_F\,, \tag{7.70}$$

where $\|\boldsymbol{C}\|_F$ is the Frobenius norm of matrix $\boldsymbol{C}$,

$$\begin{aligned}\|\boldsymbol{C}\|_F &= \sqrt{\operatorname{tr}\left(\boldsymbol{C}\boldsymbol{C}^{\mathsf{T}}\right)} && (7.71)\\ &= \sqrt{\operatorname{vec}\left(\boldsymbol{A}\right)^{\mathsf{T}}\operatorname{vec}\left(\boldsymbol{A}\right)}\,. && (7.72)\end{aligned}$$

In section 7.5.2, we log-log plot the error between a simulated estimate of the true covariance matrix and the asymptotic covariance matrix with respect to sample size. The errors approximately follow a straight line, suggesting a power law error. We fit a power function of the form an^b to the differences with respect to sample size n and found the best fit for the relative difference to be



approximately proportional to $\frac{1}{n}$ and the best fit for the absolute difference to be approximately proportional to $\frac{1}{n^2}$.

The asymptotic sampling distribution of $\hat{\boldsymbol{\theta}}(n \mid w)$ can be transformed into a Chi-squared distribution, which is given by

$$n(\hat{\boldsymbol{\theta}}(n \mid w) - \boldsymbol{\theta}^\star)^\mathsf{T} \boldsymbol{\mathcal{I}}(\boldsymbol{\theta}^\star)(\hat{\boldsymbol{\theta}}(n \mid w) - \boldsymbol{\theta}^\star) \sim \chi^2_{m \cdot q} \,. \tag{7.73}$$

Therefore, for a given n, we can construct a $(1-\alpha)$ confidence region of this statistic. We expect that $(1-\alpha)$ of the $\hat{\boldsymbol{\theta}}(n \mid w)$ point estimates to be inside of this region.

Bibliography

[1] Peter Bickel and Kjell Doksum. *Mathematical Statistics*, volume 1, chapter 2, page 117. Prentice Hall, 2 edition, 2000.

A Alternative proofs

A.1 Equation 3.6

Equation (3.6) on page 8 asserts that

$$f_{\mathrm{K,S}}(k, t \mid \boldsymbol{\Theta}^\star) = f_k(t \mid \boldsymbol{\Theta}^\star_k) \prod_{\substack{j=1 \\ j \neq k}}^{m} R_j(t \mid \boldsymbol{\Theta}^\star_j) \,.$$

Proof. First, note that $f_{\mathrm{K,S}}(k, t \mid \boldsymbol{\Theta}^\star)$ is a proper density since

$$\int_0^\infty \sum_{j=1}^{m} f_{\mathrm{K,S}}(j, t \mid \boldsymbol{\Theta}^\star)\, \mathrm{d}t = \int_0^\infty f_{\mathrm{S}}(t \mid \boldsymbol{\Theta}^\star)\, \mathrm{d}t = 1 \,. \tag{a}$$

Consider a 3-out-of-3 system. By assumption 2, T_1, T_2, and T_3 are mutually independent. Thus, the joint density that $\mathrm{T}_1 = t_1$, $\mathrm{T}_2 = t_2$, and $\mathrm{T}_3 = t_3$ is given by

$$f_{\mathrm{T}_1,\mathrm{T}_2,\mathrm{T}_3}(t_1, t_2, t_3 \mid \boldsymbol{\Theta}^\star) = \prod_{p=1}^{3} f_p(t_p \mid \boldsymbol{\Theta}_p^\star)\,. \tag{b}$$

Component 1 causes a system failure during the interval $(0, t)$, $t > 0$, if component 1 fails during the given interval and components 2 and 3 survive longer than component 1. That is, $0 < \mathrm{T}_1 < t$, $\mathrm{T}_1 < \mathrm{T}_2$, and $\mathrm{T}_1 < \mathrm{T}_3$.

Let $\Pr(t)$ denote the probability that component 1 causes a system failure during the interval $(0, t)$,

$$\Pr(t) = \Pr\left[0 < \mathrm{T}_1 < t \cap \mathrm{T}_1 < \mathrm{T}_2 \cap \mathrm{T}_1 < \mathrm{T}_3\right]\,. \tag{c}$$

This probability is given by

$$\Pr(t) = \int_{t_1=0}^{t_1=t} \int_{t_2=t_1}^{t_2=\infty} \int_{t_3=t_1}^{t_3=\infty} \prod_{p=1}^{3} f_p(t_p \mid \boldsymbol{\Theta}_p^\star)\, \mathrm{d}t_3\, \mathrm{d}t_2\, \mathrm{d}t_1\,. \tag{d}$$

Performing the integration over t_3 results in

$$\Pr(t) = \int_{t_1=0}^{t_1=t} \int_{t_2=t_1}^{t_2=\infty} \prod_{p=1}^{2} f_p(t_p \mid \boldsymbol{\Theta}_p^\star) R_3(t_1 \mid \boldsymbol{\Theta}_3^\star)\, \mathrm{d}t_2\, \mathrm{d}t_1\,. \tag{e}$$

Performing the integration over t_2 results in

$$\Pr(t) = \int_{t_1=0}^{t_1=t} f_1(t_1 \mid \boldsymbol{\Theta}_1^\star) \prod_{p=2}^{3} R_p(t_1 \mid \boldsymbol{\Theta}_p^\star)\, \mathrm{d}t_1\,. \tag{f}$$

Taking the derivative of $\Pr(t)$ with respect to t produces the density of probability near t. By the Second Fundamental Theorem of Calculus,

$$\frac{\mathrm{d}\Pr(t)}{\mathrm{d}t} = f_1(t \mid \boldsymbol{\Theta}_1^\star) \prod_{p=2}^{3} R_p(t \mid \boldsymbol{\Theta}_p^\star)\,. \tag{g}$$

The probability density $\frac{\mathrm{d}\Pr(t)}{\mathrm{d}t}$ is equivalent to the joint density that $\mathrm{K} = 1$ and $\mathrm{S} = t$ as given by theorem 3.4, i.e., $\frac{\mathrm{d}P}{\mathrm{d}t} = f_{\mathrm{K},\mathrm{S}}(1, t \mid \boldsymbol{\Theta}^\star)$. Generalizing from this completes the proof. $\square$

A.2 Equation ??

By **??**, S and W are independent and the marginal distribution of W is independent of $\boldsymbol{\theta}^\star$, thus the likelihood of observing a particular realization, $\mathbf{M_n} = \mathbf{m_n}$, is given by

$$\begin{aligned} f_{\mathbf{M_n}}(\mathbf{m_n} \mid \boldsymbol{\theta}^\star) &= \prod_{i=1}^{n} f_{\mathcal{C},\mathrm{S}\mid\mathrm{W}}\Big(\mathsf{C}_i, \mathrm{t_i} \;\Big|\; |\mathsf{C}_\mathrm{i}|, \boldsymbol{\theta}^\star\Big)\, p_{\mathrm{W}}(|\mathsf{C}_i|) \\ &= \left[\prod_{i=1}^{n} f_{\mathcal{C},\mathrm{S}\mid\mathrm{W}}\Big(\mathsf{C}_i, \mathrm{t_i} \;\Big|\; |\mathsf{C}_\mathrm{i}|, \boldsymbol{\theta}^\star\Big)\right] \left[\prod_{i=1}^{n} p_{\mathrm{W}}(|\mathsf{C}_i|)\right]. \end{aligned} \tag{A.1}$$

If we fix $\mathbf{m_n}$ and allow the parameter $\boldsymbol{\theta}$ to change, we have the likelihood function

$$\mathcal{L}\left(\boldsymbol{\theta}\,|\,\mathbf{m_n}\right) = c\prod_{i=1}^{n} f_{\mathcal{C},\mathrm{S}\,|\,\mathrm{W}}\Big(\mathsf{C}_i, \mathrm{t_i}\,\Big|\,|\mathsf{C}_\mathrm{i}|, \boldsymbol{\theta}^\star\Big)\ , \tag{A.2}$$

where the product over $p_\mathrm{W}(\ \cdot\ )$ is constant with respect to $\boldsymbol{\theta}$ and has been relabeled as c. The log-likelihood with respect to $\boldsymbol{\theta}$ then is given by

$$\begin{aligned}\ell(\boldsymbol{\theta}\,|\,\mathbf{m_n}) &= \ln c + \sum_{i=1}^{n} \ln f_{\mathcal{C},\mathrm{S}\,|\,\mathrm{W}}(\mathsf{C}_i, \mathrm{t_i}\,\big|\,|\mathsf{C}_\mathrm{i}|, \boldsymbol{\theta}) \\ &= c' + \sum_{w=1}^{m-1}\left[\sum_{i\in\mathbb{A}(w)} \ln f_{\mathcal{C},\mathrm{S}\,|\,\mathrm{W}}(\mathsf{C}_i, \mathrm{t_i}\,|\,\mathrm{w}, \boldsymbol{\theta})\right]\ ,\end{aligned} \tag{A.3}$$

where $\mathbb{A}(w) = \{i \in \{1,\,\ldots\,, n\} \colon |\mathsf{C}_i| = w\}$. Let $\mathbf{m_{n_w}} = \{(\mathsf{C}_i, \mathrm{t_i}) \colon \mathrm{i} \in \mathbb{A}(\mathrm{w})\}$. The part in brackets is the log-likelihood $\ell(\boldsymbol{\theta}\,|\,\mathbf{m_{n_w}})$. Performing this substitution yields

$$\ell(\boldsymbol{\theta}\,|\,\mathbf{m_n}) = \ln c + \sum_{w=1}^{m-1} \ell(\boldsymbol{\theta}\,|\,\mathbf{m_{n_w}})\,. \tag{A.4}$$

The score function is given by

$$\mathrm{s}\left(\boldsymbol{\theta}\,|\,\mathbf{m_n}\right) = \sum_{w=1}^{m-1} \mathrm{s}\left(\boldsymbol{\theta}\,|\,w, \mathbf{m_{n_w}}\right)\ , \tag{A.5}$$

The information matrix may be computed by taking the negative of the expectation of the Jacobian of the score over the true joint distribution of S, $\mathcal{C}$, and W, giving

$$\boldsymbol{\mathcal{I}}\left(\boldsymbol{\theta}^\star\right) = -\,\mathbb{E}_{\boldsymbol{\theta}^\star}\left[\mathcal{J}\left(\mathrm{s}\left(\boldsymbol{\theta}\,|\,\mathbf{M_1}\right)\right)\Big|_{\boldsymbol{\theta}=\boldsymbol{\theta}^\star}\right]\,. \tag{A.6}$$

Substituting eq. (A.5) with its definition gives

$$\boldsymbol{\mathcal{I}}\left(\boldsymbol{\theta}^\star\right) = -\,\mathbb{E}_{\boldsymbol{\theta}^\star}\left[\mathcal{J}\left(\sum_{w=1}^{m-1}\mathrm{s}\left(\boldsymbol{\theta}\,|\,w, \mathbf{M}(w,1)\right)\right)\Bigg|_{\boldsymbol{\theta}=\boldsymbol{\theta}^\star}\right]\,. \tag{A.7}$$

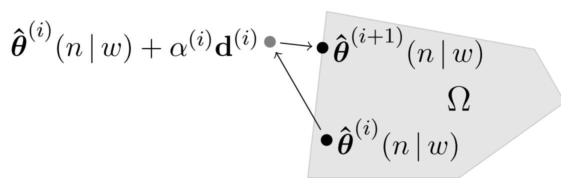
The Jacobian is a linear operator so it can be moved inside the summation, giving

$$\boldsymbol{\mathcal{I}}\left(\boldsymbol{\theta}^\star\right) = -\,\mathbb{E}_{\boldsymbol{\theta}^\star}\left[\sum_{w=1}^{m-1}\mathcal{J}\left(\mathrm{s}\left(\boldsymbol{\theta}\,|\,w, \mathbf{M}(w,1)\right)\right)\Bigg|_{\boldsymbol{\theta}=\boldsymbol{\theta}^\star}\right]\,. \tag{A.8}$$

The Jacobian of s is the Hessian of ℓ. Performing this substitution gives

$$\boldsymbol{\mathcal{I}}\left(\boldsymbol{\theta}^\star\right) = -\,\mathbb{E}_{\boldsymbol{\theta}^\star}\left[\sum_{w=1}^{m-1}\mathcal{H}\left(\ell\left(\boldsymbol{\theta}\,|\,w, \mathbf{M}(w,1)\right)\right)\Bigg|_{\boldsymbol{\theta}=\boldsymbol{\theta}^\star}\right]\,. \tag{A.9}$$

Figure 3: Projection onto convex set Ω.



The expectation is a linear operator so it may be moved inside the summation, giving

$$\boldsymbol{\mathcal{I}}\left(\boldsymbol{\theta}^{\star}\right) = -\sum_{w=1}^{m-1} \mathbb{E}_{\boldsymbol{\theta}^{\star}}\left[\mathcal{H}\left(\ell\left(\boldsymbol{\theta} \mid w, \mathbf{M}(w,1)\right)\right)\Big|_{\boldsymbol{\theta}=\boldsymbol{\theta}^{\star}} \right]. \tag{A.10}$$

The expectation sums over the probability mass function $\mathrm{W} \sim p_{\mathrm{W}}(\,\cdot\,)$, giving

$$\boldsymbol{\mathcal{I}}\left(\boldsymbol{\theta}^{\star}\right) = -\sum_{w=1}^{m-1} p_{\mathrm{W}}(w)\, \mathbb{E}_{\boldsymbol{\theta}^{\star}}\left[\mathcal{H}\left(\ell\left(\boldsymbol{\theta} \mid w, \mathbf{M}(w,1)\right)\right) \right], \tag{A.11}$$

where the expectation is now over the true marginal joint distribution of $\mathcal{C}$ and S given $\mathrm{W} = w$. The expectation of the Hessian of ℓ is equivalent to the negative of $\boldsymbol{\mathcal{I}}(\boldsymbol{\theta}^{\star} \mid w, \mathbf{M}(w,1))$. Performing this substitution gives

$$\boldsymbol{\mathcal{I}}\left(\boldsymbol{\theta}^{\star}\right) = \sum_{w=1}^{m-1} p_{\mathrm{W}}(w)\, \boldsymbol{\mathcal{I}}\left(\boldsymbol{\theta}^{\star} \mid w\right). \tag{A.12}$$

$\square$

B Numerical solutions to the MLE

The function $\ell(\boldsymbol{\theta} \mid w, \mathbf{m_n})$ is the log-likelihood with respect to $\boldsymbol{\theta} \in \Omega$ where $\Omega \subset \mathbb{R}^{m \cdot q}$. This function has a surface in an $(m \cdot q + 1)$ dimensional space, where a particular point on this surface represents the log-likelihood of observing $\mathbf{m_n}$ with respect to $\boldsymbol{\theta}$. By definition 5.3, $\hat{\boldsymbol{\theta}}(n \mid w)$ is the point on this surface that is at a maximum,

$$\hat{\boldsymbol{\theta}}(n \mid w) = \arg\max_{\boldsymbol{\theta} \in \Omega} \ell(\boldsymbol{\theta} \mid w, \mathbf{m_n}). \tag{5.9 revisited}$$

Generally there is no closed-form solution that solves $\hat{\boldsymbol{\theta}}(n \mid w)$, in which case iterative search methods may be used to numerically approximate a solution.

The general version of iterative search that numerically approximates a solution to eq. (5.10), subject to the constraint $\boldsymbol{\theta}^{\star} \in \Omega$, is shown in algorithm 5. Since iterative search is a local search method, it may fail to converge to a global maximum.

Assume parameter space Ω is convex. Then, the function `Project`$(\boldsymbol{\theta}, \Omega)$ in algorithm 5 projects any point $\boldsymbol{\theta}$ to the nearest point in the parameter space Ω as depicted by fig. 3. Thus, the search method is restricted to searching over the feasible parameter space.

In algorithm 5, $\mathbf{d}^{(i)}$ is a unit vector denoting a *promising* direction in which to search for a better solution than $\hat{\boldsymbol{\theta}}^{(i)}(n \mid w)$. The Fisher scoring algorithm is a slight variation of Newton-Raphson[2] in

[2] If the *observed* information matrix is used instead of the *expected* information matrix, the Fisher scoring algortihm is equivalent to Newton-Raphson.

Algorithm 5: Iterative maximum likelihood search

Result: an approximate solution to the stationary points of the maximum likelihood equation

Input:

$\mathbf{\Theta}^{\star}$, the true parameter index.

Output:

$\hat{\boldsymbol{\theta}}(n \,|\, w)$, an approximation of the maximum likelihood estimate.

Model `FindMLE`($\mathbf{m_n}$)

  $\hat{\boldsymbol{\theta}}^{(0)}(n \,|\, w) \leftarrow$ an initial starting point in Ω

  $i \leftarrow 0$

  while stopping criteria *not satisfied* **do**

    $\hat{\boldsymbol{\theta}}^{(i+1)}(n \,|\, w) \leftarrow \texttt{Project}\left(\hat{\boldsymbol{\theta}}^{(i)}(n \,|\, w) + \alpha^{(i)}\mathbf{d^{(i)}}, \Omega\right)$

    $i \leftarrow i + 1$

  end

  return $\hat{\boldsymbol{\theta}}^{(i)}(n \,|\, w)$

which

$$\mathbf{d}^{(i)} = \boldsymbol{\mathcal{I}}^{-1}\left(\hat{\boldsymbol{\theta}}^{(i)}(n \,|\, w)\right) \mathrm{s}\left(\hat{\boldsymbol{\theta}}^{(i)}(n \,|\, w)\right) . \tag{B.1}$$

The scalar $\alpha^{(i)} > 0$ is the distance to move from point $\hat{\boldsymbol{\theta}}^{(i)}(n \,|\, w)$ to generate the next point, $\hat{\boldsymbol{\theta}}^{(i+1)}(n \,|\, w)$. Most naturally, the solution to

$$\alpha^{(i)} = \arg\max_{\alpha > 0} \ell(\hat{\boldsymbol{\theta}}^{(i)}(n \,|\, w) + \alpha \cdot \mathbf{d}^{(i)}) \tag{B.2}$$

is desired, e.g., using *Golden Section* line search to find an approximate solution. Finally, the iterations are repeated until some *stopping criteria* is satisfied. Most naturally, this is given by

$$\| \mathrm{s}(\hat{\boldsymbol{\theta}}^{(i)}(n \,|\, w)) \| < \epsilon , \tag{B.3}$$

signifying a stationary point has been reached.

C Generic joint distribution

We assume the distribution of W does not carry information about the true parameter index $\boldsymbol{\theta}^{\star}$. By the axioms of probability, the joint distribution of $\mathcal{C}$, S, and W is given by

$$\begin{aligned} f_{\mathcal{C},\mathrm{S},\mathrm{W}}(\mathsf{C}, t, w \,|\, \boldsymbol{\theta}^{\star}) = \\ p_{\mathcal{C} \,|\, \mathrm{S},\mathrm{W}}(\mathsf{C} \,|\, t, w, \boldsymbol{\theta}^{\star}) p_{\mathrm{W} \,|\, \mathrm{S}}(w \,|\, t) f_{\mathrm{S}}(t \,|\, \boldsymbol{\theta}^{\star}) \, . \end{aligned} \tag{C.1}$$

Since W carries no information about the true parameter index $\boldsymbol{\theta}^{\star}$, the maximum likelihood estimate $\hat{\boldsymbol{\theta}}_{\mathbf{n}}$ is independent of the distribution of W.

Proof.

$$
\begin{aligned}
\hat{\boldsymbol{\theta}}_{\mathbf{n}} &= \arg\max_{\boldsymbol{\theta}} \ell(\boldsymbol{\theta} \mid \mathbf{m_n}) \\
&= \arg\max_{\boldsymbol{\theta}} \Bigg[\sum_{i=1}^{n} \ln p_{\mathcal{C} \mid \mathrm{S,W}} \left(\mathsf{C}_i \mid t_i, |\mathsf{C}_i|, \boldsymbol{\theta} \right) \\
&\quad + \sum_{i=1}^{n} \ln f_{\mathrm{S}} \left(t_i \mid \boldsymbol{\theta} \right) + \underbrace{\sum_{i=1}^{n} \ln p_{\mathrm{W} \mid \mathrm{S}} \left(|\mathsf{C}_i| \mid t_i \right)}_{\text{constant}} \Bigg] ,
\end{aligned} \tag{a}
$$

where the last term is constant with respect to $\boldsymbol{\theta}$ and may be dropped from the maximum likelihood equation without changing the point that maximizes the likelihood. □

The sampling distribution of $\hat{\boldsymbol{\theta}}_{\mathbf{n}}$ is a function of $\mathbf{M_n}$ (as opposed to a particular realization), and since $\mathbf{M_n}$ is a function of W (the distribution of the cardinality of candidate sets), the sampling distribution of $\hat{\boldsymbol{\theta}}_{\mathbf{n}}$ is also a function of W.

Given a statistical model, the asymptotic sampling distribution of the maximum likelihood estimator is nearly automatic. The most taxing (and uncertain) part is the *model selection.* Since we may not have much confidence in any particular model of the joint distribution of W and S, the *expected* information matrix is sketchy. In general, the *observed* information matrix is preferred. The sampling distribution of $\hat{\boldsymbol{\theta}}_{\mathbf{n}}$ converges in distribution to a multivariate normal with a mean given by $\boldsymbol{\theta}^\star$ and a variance-covariance given by the inverse of the *observed* information matrix, written

$$
\mathbf{Y_n} \xrightarrow{d} \mathrm{MVN}\left(\boldsymbol{\theta}^\star, \boldsymbol{\mathcal{J}}_{\mathrm{n}}{}^{-1} \left(\boldsymbol{\theta}^\star \mid \mathbf{M_n} \right) \right) . \tag{C.2}
$$

If an accurate distribution of $\mathrm{W} \mid \mathrm{S} = \mathrm{t}$ could be constructed such that it is dependent on $\boldsymbol{\theta}^\star$, then the distribution of W in a sample carries extra information about $\boldsymbol{\theta}^\star$. A plausible model of W would depend upon the entropy of $\mathrm{K} \mid \mathrm{S} = \mathrm{t}$.